

# difes: the test ledger

Every check the program has passed through, graded on what it proved and against what

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# What this document is

difes is a personal project: an attempt to give the group a modernised version of what FOCUS does, so that our dielectric-function models, fits and simulations can be made in a detailed, safe, open and documented way, on the machines we use. The question that matters to someone who has used FOCUS every day is not whether difes is modern but whether it is right. This ledger answers it test by test, in the order the tests were run, from the physics identities of the first release to the FOCUS expert’s own fits. Each record states the objective as a question, the reference difes was compared against and how strong that reference is, the data and where they come from, what was done, what came out, and whether the objective was met. The records live in one file on the project site (`data/tests.yml`); this document and the site’s Tests section are generated from it.

# The scales

Grade of achievement, five levels:

| grade               | meaning                                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| A · exact           | agreement at the precision of the reference itself (the file’s digits, machine precision), nothing fitted                  |
| B · within the data | curves agree within the data’s own scatter or the digitisation precision; parameters agree where the problem is well posed |
| C · explained       | a difference exists, its origin is identified and quantified, and difes’s side is the defensible one                       |
| D · open            | a difference is observed and a hypothesis exists; an author or a file is needed to confirm it                              |

| grade       | meaning                                              |
|-------------|------------------------------------------------------|
| E · blocked | the test cannot be run yet, and the reason is stated |

**Class of reference**, four classes: I, FOCUS’s numerical output (curves exported to seven significant digits, or FOCUS’s own formulas); II, FOCUS’s output as a vector figure, read exactly from the paths of a PDF; III, printed tables and bitmap figures, digitised to about  $5 \times 10^{-3}$  of the axis; IV, physics alone (identities, limits, closed forms, independent programs). The class says how much a grade is worth: an A against class I is the strongest statement the ledger can make. A D is not a failure of difes; in every D so far the difference is between a publication’s table and its own figure, and the question is put to the author. An E is a test waiting for an input, listed so that the gaps are as visible as the results.

## The ledger in numbers

|                                 |                                                     |
|---------------------------------|-----------------------------------------------------|
| tests in the ledger             | 40, of which 36 run                                 |
| exact (A)                       | 20, 8 of them against FOCUS’s own output or figures |
| within the data (B)             | 8                                                   |
| explained (C)                   | 4                                                   |
| open, waiting for an author (D) | 4                                                   |
| blocked (E)                     | 4                                                   |
| by class of reference           | I 15 · II 5 · III 7 · IV 13                         |

## The tests, in order

### T01 · The profiles against physics: limits, closed forms, passivity, causality

**Grade A** · **exact** · class IV (physics alone) · 2026-09-29 · difes 0.1.0 · log L014

**Objective.** Do difes’s dielectric-function profiles satisfy the closed forms, limits and causality that physics imposes, before any comparison with FOCUS?

**Reference.** Closed forms: the Drude value at  $\omega = \omega_p$ , the Lorentz static limit  $\chi(0) = S$  and its value at resonance, the Lyddane–Sachs–Teller relation; the Kramers–Kronig relation between the real and imaginary parts of every term, computed numerically on a wide grid.

**Data.** Synthetic; no measured data.

**Physics covered.** Drude, Lorentz (strength form), four-parameter factorised mode, causal Gaussian; passivity  $\varepsilon'' \geq 0$ ; the high-frequency limit  $\varepsilon \rightarrow \varepsilon_\infty$ .

**Procedure.** The automated suite (pytest, one command) evaluates each profile and checks the closed forms exactly, passivity on the whole grid, that the four-parameter mode reduces to a Lorentz oscillator when  $\gamma_{TO} = \gamma_{LO}$  with  $\omega_p^2 = \varepsilon_\infty(\omega_{LO}^2 - \omega_{TO}^2)$ , the Gaussian’s peak, FWHM and even symmetry to  $10^{-6}$ , and for each term that  $\text{Re } \chi$  is the Kramers–Kronig transform of  $\text{Im } \chi$  (Maclaurin rule on a grid to  $40\,000\text{ cm}^{-1}$ ).

**Result.** Every check holds: closed forms to machine precision, causality to 1 % of the scale for Drude, Lorentz and the four-parameter mode and to 0.1 % for the causal Gaussian, whose published formula the knowledge pages had flagged and this test verified. 16 test functions (18 cases) in tests/test\_models.py; the whole suite has 134 cases at 0.16.0 and passes at every release.

**Verdict.** Achieved. The floor under everything else: difes agrees with physics before it is asked to agree with FOCUS.

*Where to look:* difes/tests/test\_models.py, conftest.py (the causality oracle); Models of the dielectric function (theory).

## T02 · The optics against textbook values: Fresnel, slabs, films, energy balance

**Grade A** · **exact** · class IV (physics alone) · 2026-09-29 · difes 0.1.0 – 0.3.0 · log L014

**Objective.** Do the reflectance, transmittance and emissivity of bulk samples, slabs and films obey the Fresnel equations, energy conservation and their limiting cases?

**Reference.** Textbook values ( $R = 0.04$  for  $n = 1.5$ ,  $R = 1$  for a lossless metal),  $R + T + E = 1$ , the lossless-slab resonances, the opaque and thin limits, and FOCUS's library formula  $\text{Emittance-E} = (1 - R)(1 - T)/(1 - RT)$ .

**Data.** Synthetic dielectric functions.

**Physics covered.** Principal root of  $\varepsilon$ ; normal-incidence Fresnel; coherent and incoherent slabs; film on a substrate in both coherence modes; the conversions  $\varepsilon \rightarrow (n, k)$  ( $r, \theta$ ).

**Procedure.** 12 tests: branch choice of the index; textbook reflectances; conversion round trips; lossless-slab limits ( $T = 1$  at  $\theta = 0$ , incoherent  $R + T = 1$ );  $R + T + E = 1$  for an absorbing plate in both modes; a thick absorbing slab tends to the opaque reflectance ( $10^{-9}$ ); a film with air as substrate reproduces the free slab, with zero thickness the bare substrate, with large thickness its own bulk; the slab emissivity equals FOCUS's Emittance-E for three thicknesses.

**Result.** All hold, to  $10^{-12}$  where the identity is exact and  $10^{-9}$  in the limits.

**Verdict.** Achieved.

*Where to look:* difes/tests/test\_optics.py; From  $\varepsilon$  to the measured spectrum (theory).

## T03 · Kramers–Kronig, MSKK and CFRC against analytic phases and an external oracle

**Grade A** · **exact** · class IV (physics alone) · 2026-09-29 · difes 0.4.0 · log L014

**Objective.** Does the finite-range Kramers–Kronig machinery recover known phases and dielectric functions, and do the two corrections improve on the plain transform as the theory says?

**Reference.** The analytic phase and  $\chi'$  of a Lorentz oscillator; SciPy's Cauchy-weight quadrature on the same interpolant; a fixture computed by the independent OptiCFRC program (Lorentz  $\varepsilon_\infty = 3$ ,  $\omega_0 = 500$ ,  $\omega_p = 800$ ,  $\gamma = 25$  on  $100\text{--}2000\text{ cm}^{-1}$ ).

**Data.** Synthetic reflectivities; the OptiCFRC fixture file of 300 rows.

**Physics covered.** Principal-value kernel in closed form (linear and cubic interpolants); KK phase in direct and derivative forms; multiply-subtractive KK with anchors; CFRC as phase-log terms with poles, as a positivity constraint, and as a  $\chi$  extension with knots.

**Procedure.** 14 tests: the closed-form kernel against quadrature ( $10^{-4}$ );  $\chi'$  from  $\chi''$  and the phase from  $R$  recover the analytic Lorentz on a wide range ( $10^{-3}$ ); the truncation error is shown to exceed 0.02 on a narrow range; against the OptiCFRC fixture the phase agrees to  $5 \times 10^{-3}$  and  $\chi'$  to  $2 \times 10^{-3}$  of its maximum; MSKK hits its anchors to  $10^{-9}$  and lowers the error; CFRC beats MSKK, which beats plain KK, on the quartz-like two-mode case of the CFRC paper; the positivity constraint alone recovers a phase that touches zero.

**Result.** All hold with the stated tolerances; 2000 points transform in under 2 s.

**Verdict.** Achieved against physics and against one independent program. The exact test against FOCUS's own KK output is still to come (an export with its KK curve).

*Where to look:* difes/tests/test\_kk.py; tests/fixtures/opticfrc\_kk\_lorentz.txt; Kramers–Kronig on a finite range (theory).

## T04 · The fitter on synthetic truth: recovery, errors, bounds, linking, Jacobians

**Grade A** · **exact** · class IV (physics alone) · 2026-09-29 · difes 0.2.0 · log L014

**Objective.** Started from a wrong guess, does the fitter return the parameters that generated a spectrum, with honest uncertainties, respecting frozen values, bounds and parameters shared between spectra?

**Reference.** The two-Lorentz model of the RefFIT manual's Fig. 3-11 ( $\varepsilon_\infty$  5.143; 474.65/255.79/14.097; 238.68/1074.2/8.6534) as the truth; central finite differences for the analytic Jacobian.

**Data.** Synthetic R, E,  $\varepsilon'$  and  $\varepsilon''$  spectra with known noise.

**Physics covered.** Least squares with analytic derivatives for Drude, Lorentz and Gaussian terms, finite differences for the rest; weighting by  $\sigma$ ; linked parameters across a temperature series.

**Procedure.** 11 tests: all seven parameters recovered within  $3.5\sigma$  and 0.5 % with  $0.5 < \chi^2_{\text{red}} < 2$ ; frozen parameters stay and carry no error; a bound that excludes the truth is respected; a two-Drude metal with one damping frozen; two spectra sharing one  $\varepsilon_\infty$ ; the analytic Jacobian against finite differences to  $10^{-4}$  for four quantities; a weighted joint fit of R and E.

**Result.** All hold.

**Verdict.** Achieved. The same fitter later returned FOCUS's parameters from a perturbed start on FOCUS's own projects (T16, T17).

*Where to look:* difes/tests/test\_fit.py; Least-squares fitting (theory).

## T05 · The software around the physics: language, spectra, sessions, server, Bench

**Grade A** · **exact** · class IV (physics alone) · 2026-09-30 · difes 0.5.0 – 0.11.0 · log L014

**Objective.** Do the parts a user touches behave as documented: the math language, the spectrum readers, the session graph, the command line, the local server and the Bench recipes?

**Reference.** The documented behaviour, FOCUS's own formula-editor examples, and the sessions built from Irati's data.

**Data.** Synthetic spectra; FOCUS's help examples; Irati's chromium spectrum and sessions.

**Physics covered.** Where a test fits or inverts, the synthetic truth is recovered ( $\omega_0 = 500$  to 0.2 %; the KK inversion of R returns  $\varepsilon$ ).

**Procedure.** 9 tests of the math language (precedence, complex constants, FOCUS's five worked examples, user functions, the sandbox, an expression term that equals the built-in Lorentz and fits); 12 of spectra (units, reversal of wavelength axes, column selection, Fortran exponents, round trips, bounds); 9 of the session graph (addresses, caching and invalidation, views, JSON, errors, MSKK and CFRC as nodes, libraries and scopes); 6 of the command line; 9 of the server (API, PNG rendering, OSML import of the quartz model, multi-tab sessions, the Bench routes); 6 of the Bench (tagging, reading Irati's sessions, suggestions that find two oscillators, the staged guided fit with adopt and discard, inversion).

**Result.** 51 tests pass; the suite runs in 12 s.

**Verdict.** Achieved.

*Where to look:* difes/tests/test\_expr.py, test\_spectra.py, test\_graph.py, test\_cli.py, test\_server.py, test\_bench.py; Python reference (manual).

## T06 · FOCUS's OSML files: quartz E, LiNbO<sub>3</sub> E, the four libraries, the round trip

**Grade A** · **exact** · class I (FOCUS's numerical output) · 2026-09-30 · difes 0.6.0 – 0.7.0 · log L023, L025

**Objective.** Does difes read FOCUS's own model files and function libraries and compute from them exactly what its native model classes and optics compute?

**Reference.** FOCUS 2.0's distribution: the OSML example models quartz E ( $\varepsilon_\infty$  2.38584, eight four-parameter modes) and LiNbO<sub>3</sub> E, and its four OSML libraries (Fresnel coefficients at oblique incidence, reflectance, transmittance and emittance of plates).

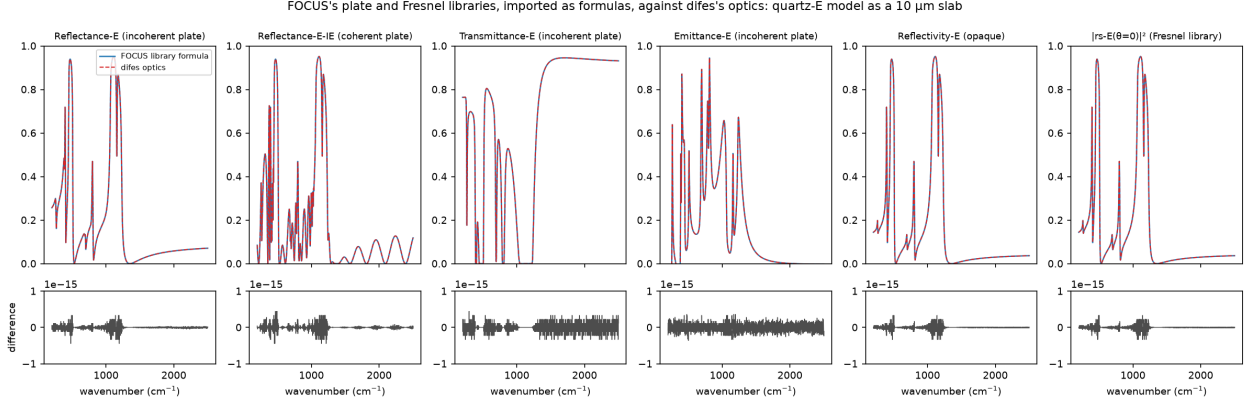
**Data.** FOCUS's files as shipped, under focus\_reference/; evaluated on grids of 1–5000 cm<sup>-1</sup>. No measured spectrum.

**Physics covered.** Four-parameter product with  $\varepsilon_\infty$ ; incoherent and coherent plate formulas; oblique-incidence Fresnel coefficients.

**Procedure.** Import each model with difes.osml; evaluate the imported model and the file's own formula on the same grid; export the model back to OSML and import it again; evaluate each library formula and difes's optics function at the same  $\varepsilon$  and thickness.

**Result.** Quartz E and LiNbO<sub>3</sub> E import unchanged; the round trip gives  $\max |\Delta R| = 0$ ; the four libraries agree with difes's optics to  $4 \times 10^{-16}$ . The quartz file exposed one correction: the index must take the principal root ( $n \geq 0$ ), because the four-parameter product is slightly non-passive between modes.

**Verdict.** Achieved. FOCUS's model algebra and its plate optics are the ones difes implements.



**Figure 1:** FOCUS's four library formulas evaluated in difes against difes's native optics functions, and the OSML round trip.

Where to look: Exchanging with FOCUS (manual); difes/examples/osml\_figures.py; tests/test\_osml.py.

## T07 · FOCUS's silica example: anharmonic phonons and Voigt, read from the help images

**Grade B** · within the data · class III (printed tables and bitmap figures) · 2026-09-30 · difes 0.7.0 · log L025

**Objective.** Can the two profiles FOCUS documents only as pictures (phonon-anharmonic-i, Voigt) be implemented so that FOCUS's silica example computes FOCUS's plotted emittance?

**Reference.** The FOCUS help: the images of the two definitions, and the plotted emittance of the 161  $\mu\text{m}$  silica plate (a 250-pixel figure with guessed axis limits).

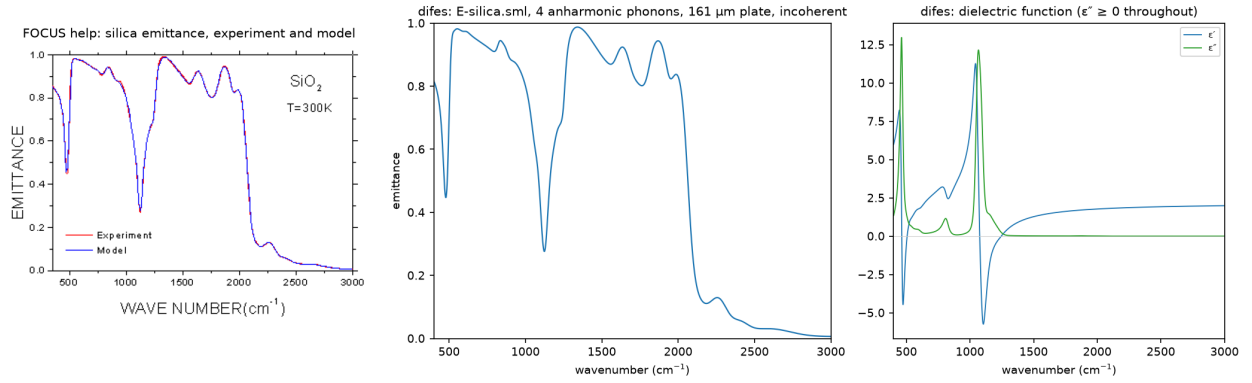
**Data.** FOCUS's OSML silica model; the help figure read by eye at its features (minima, edge, bump).

**Physics covered.**  $\Delta\varepsilon \Omega^2 / (\Omega^2 - x^2 + i\Gamma)$  with  $\Gamma$  a self-energy built from Gaussian and Voigt peaks; incoherent plate emittance.

**Procedure.** Implement the two profiles in the math language with difes's sign convention ( $\varepsilon'' \geq 0$ ); import the silica model; compute the plate emittance; compare the features with the help figure.

**Result.** Feature for feature: minima 0.45 at 480  $\text{cm}^{-1}$  and 0.28 at 1120  $\text{cm}^{-1}$ , the edge at 2000  $\text{cm}^{-1}$ , the bump at 2250  $\text{cm}^{-1}$ . The agreement also confirms that FOCUS's  $+i\Gamma$  is the opposite time convention with the same reflectivity.

**Verdict.** Achieved at the level a small picture allows: the shapes are FOCUS's. The exact level for these profiles came later with the sample projects (T18, T19).



**Figure 2:** FOCUS’s help plot of the silica plate emittance beside difes’s recomputation of the same OSML model.

*Where to look:* Models of the dielectric function (theory).

## T08 · Irati’s chromium: forward optics, and FOCUS’s film formula

**Grade C · explained · class I** (FOCUS’s numerical output) · 2026-10-01 · difes 0.14.0 · log L039, L041

**Objective.** With the same parameters, does difes compute the same reflectance as FOCUS’s project files for the bare metal and for the oxide film?

**Reference.** Irati Tolosa’s two FOCUS projects exported as OSML (Bestebat.sml, Azkena.sml, TFG 2025), whose formulas FOCUS computes; the report’s figure 20 as FOCUS’s exported curve (curve.dat).

**Data.** The projects’ definitions evaluated on 400–1350  $\text{cm}^{-1}$ ; the report’s Tables 4 and 6 (fourteen Lorentz oscillators, one  $\omega_0$  not transcribed, thickness not stated); FOCUS’s drawn curve of figure 20.

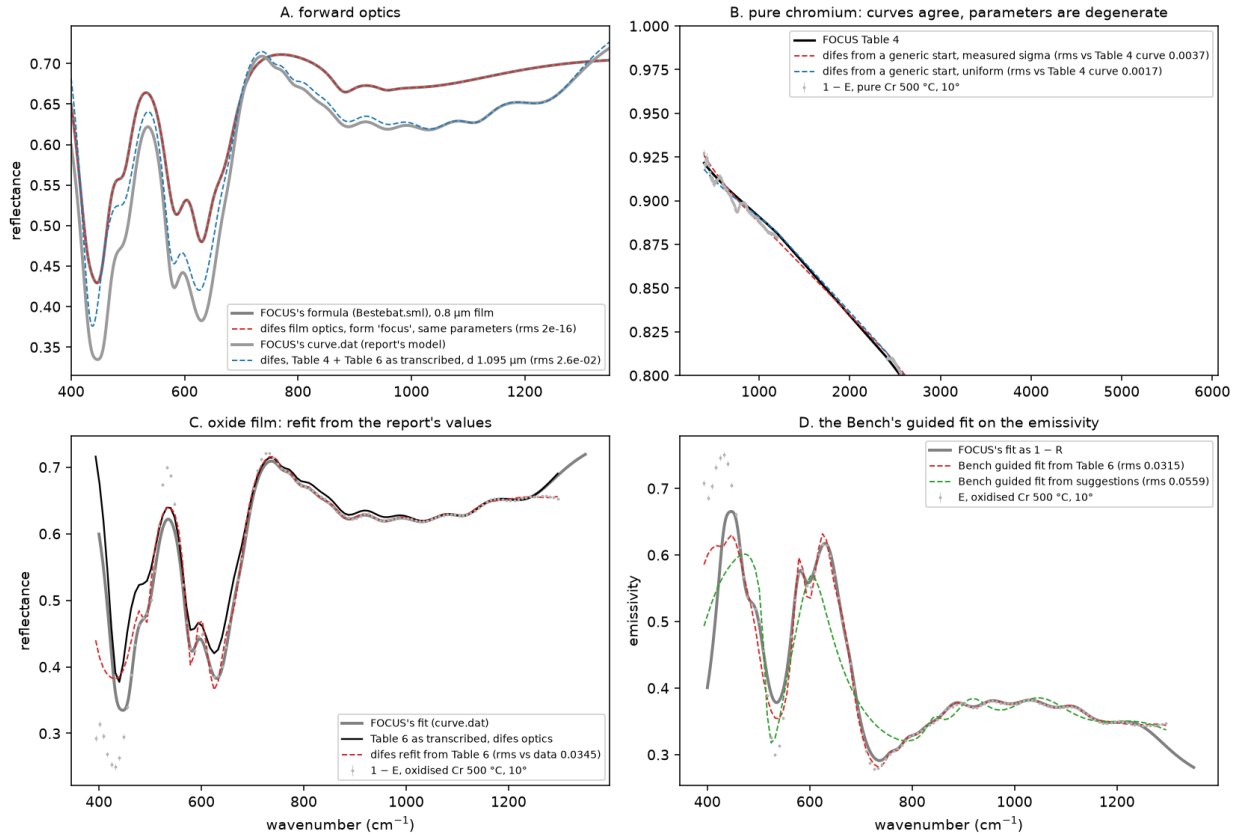
**Physics covered.**  $\varepsilon_\infty$  plus two Drude terms for chromium; eight phonons for the oxide; an incoherent 0.8  $\mu\text{m}$  film on the metal, FOCUS’s eq. 27 (Reflectance-Ei-E1-E2).

**Procedure.** Evaluate FOCUS’s formulas in the math language and difes’s native sample optics at identical parameters; compare the two film formulas; fit only the missing  $\omega_0$  and the thickness against curve.dat under each formula.

**Result.** Bare metal  $1.5 \times 10^{-16}$ ; the film with the FOCUS form  $1.7 \times 10^{-16}$ ; the film with difes’s default energy-conserving form  $8.8 \times 10^{-3}$  rms, 0.049 at most, inside the reststrahlen band of  $\text{Cr}_2\text{O}_3$ . FOCUS writes the incoherent film’s numerator as  $|(1 - r_1^2) r_2 \tau^2|^2$ , difes as  $(1 - R_1)^2 R_2 A$ ; they coincide for a transparent film and differ where it absorbs.

**Verdict.** Explained and absorbed: difes 0.14.0 offers both forms (form=“focus” on the sample, the formula selector of the Bench), so FOCUS’s curve is reproduced to machine precision, and the manual’s earlier claim that the two agree to 1 % is withdrawn for absorbing films.

**Open point.** Which film symbol drew the report’s figure 20 is a curiosity of the report: the energy form fits curve.dat better ( $9.9 \times 10^{-3}$ ) than the form the later files use ( $1.8 \times 10^{-2}$ ).



**Figure 3:** Panel A: FOCUS's film formula and difes's film optics with the FOCUS form coincide; the report's model against FOCUS's drawn curve.

Where to look: Validation against FOCUS, level A; difes/examples/reproduce\_irati.py.

## T09 · Irati's chromium: the pure-metal fit

**Grade B · within the data ·** class I (FOCUS's numerical output) · 2026-10-01 · difes 0.14.0 · log L041

**Objective.** Does difes's fitter find FOCUS's curve for the pure chromium, and its parameters, from the report's values and from generic starts?

**Reference.** The report's Table 4 ( $\varepsilon_\infty$  4.86, Drude (15344, 834) and (26930, 14258)) and the curve it produces; Irati's measured emissivity with its uncertainties.

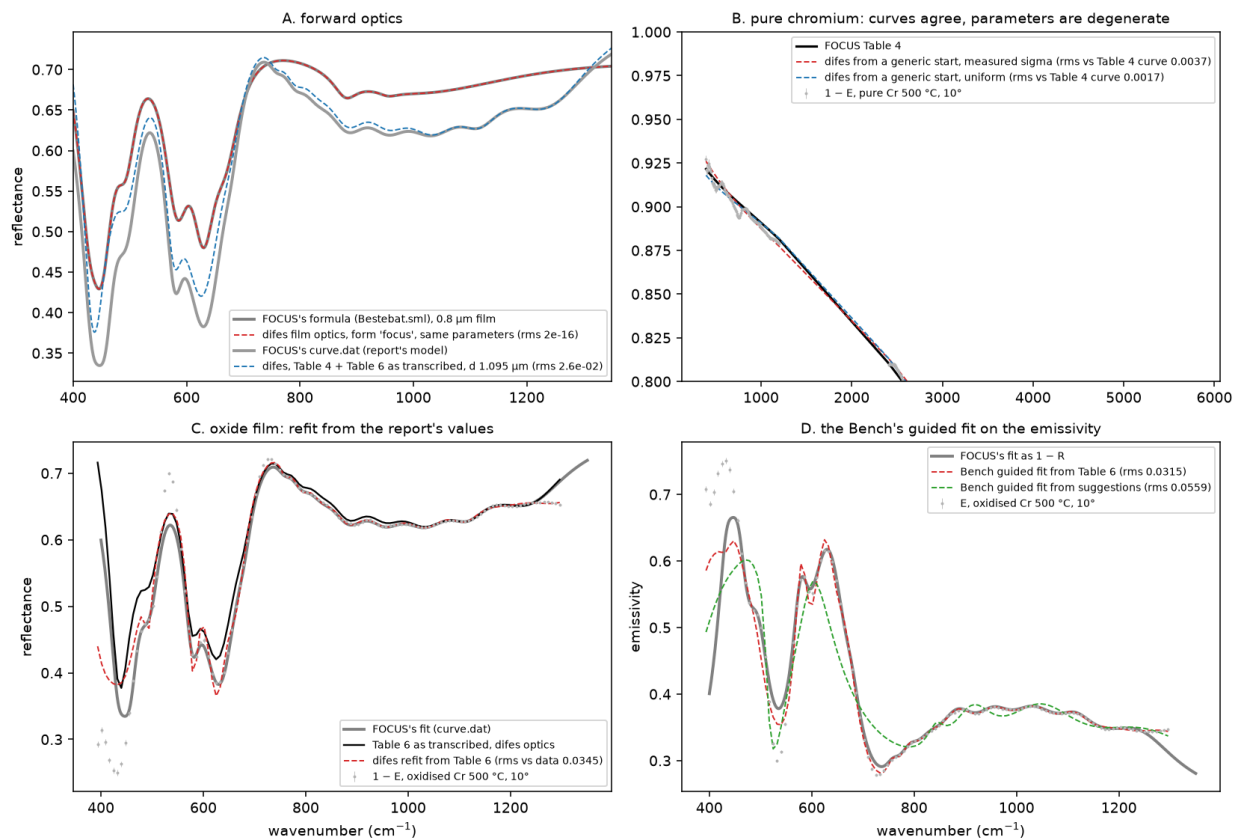
**Data.** The 500 °C, 10° emissivity of the pure sample, 545 points after excluding the 1200–2400  $\text{cm}^{-1}$  dip the report attributes to the measurement; fitted as  $R = 1 - E$ .

**Physics covered.**  $\varepsilon_\infty$  plus two Drude terms; a metal's reflectance near 0.95.

**Procedure.** Fit from Table 4, from Table 4  $\times$  0.7 and from a generic start, under the measured  $\sigma$  and under uniform weights; compare curves with the Table 4 curve and with the data; repeat with  $\varepsilon_\infty$  frozen at 4.86.

**Result.** All starts converge to one answer per weighting; the curves lie 0.0017–0.0037 from the Table 4 curve, below the data's scatter (median  $\sigma$  0.0015; Table 4's own residual 0.0075); difes's residual 0.0069–0.0071 is lower than Table 4's. The parameters are not Table 4's and depend on the weighting, the second Drude term running to its bounds.

**Verdict.** Achieved for the curve. The parameters are degenerate: the reflectance fixes a combination of  $\varepsilon_\infty$  and the plasma frequencies, not each of them, as the report itself says of its second oscillator. Reproducing Table 4's numbers would need Irati's unrecorded stopping choices.



**Figure 4:** Panel B: pure chromium, the Table 4 curve and difes's fits from a generic start under two weightings.

Where to look: Validation against FOCUS, level B.

## T10 · Irati's chromium: the oxide film refit and the Bench's way

**Grade B · within the data · class I** (FOCUS's numerical output) · 2026-10-01 · difes 0.14.0 – shell 0.7.1 · log L041

**Objective.** Refitting the report's fourteen-oscillator oxide film, does difes reach FOCUS's curve and keep the report's band positions? Can a user get there through the Bench's guided fit?

**Reference.** Table 6 of the report; FOCUS's fitted curve curve.dat; the measured emissivity of the oxidised sample.

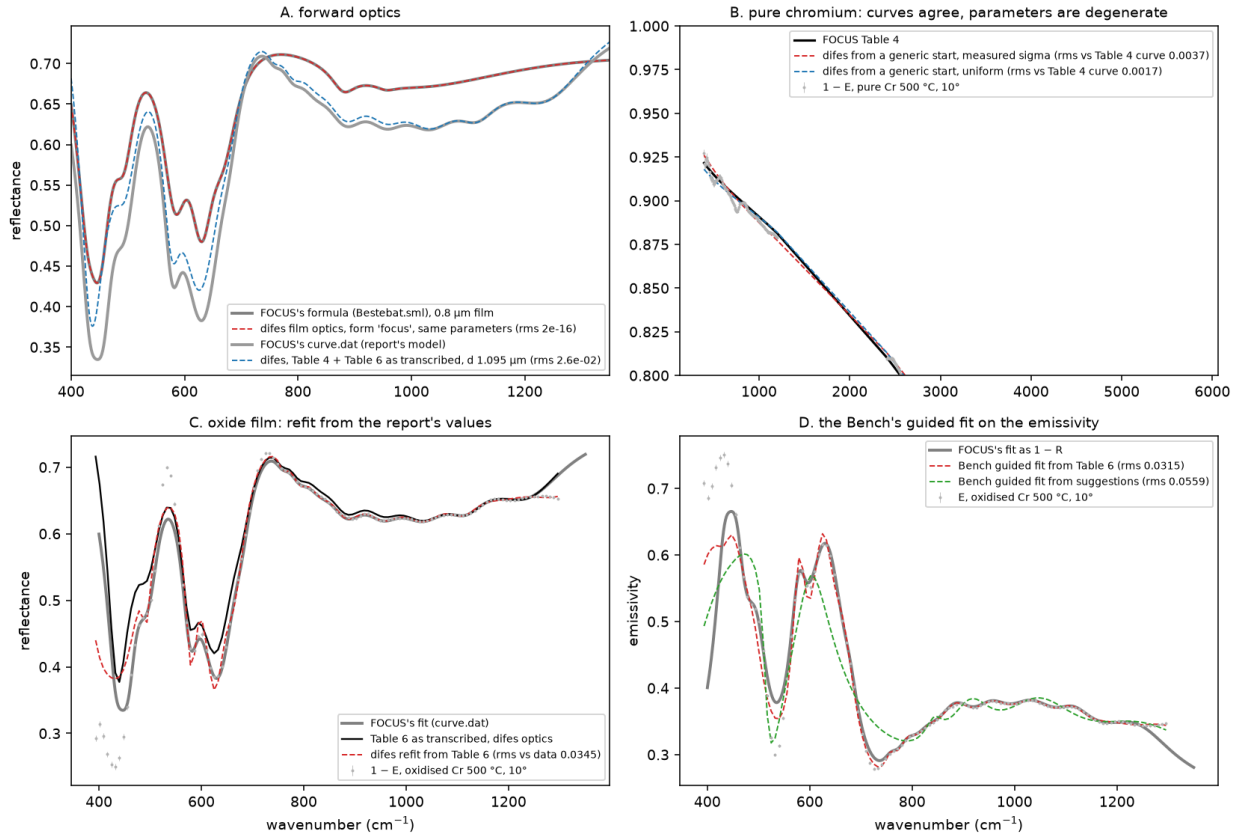
**Data.** 118 points on 390–1300  $\text{cm}^{-1}$ , the report's fit range; a 1.095  $\mu\text{m}$  incoherent film with the FOCUS form on the frozen Table 4 chromium.

**Physics covered.** Fourteen Lorentz oscillators of  $\text{Cr}_2\text{O}_3$  on a metal; incoherent film optics, FOCUS form.

**Procedure.** Refit from Table 6 and from a perturbed Table 6 under both weightings; count oscillators that stay within their own width of the report's positions; then the Bench recipe: import, model, Film with the FOCUS formula, Guided fit in four stages, adopt, from Table 6 and from the Bench's own suggestions.

**Result.** Table 6 as transcribed sits 0.074 from the data and FOCUS's own curve 0.046; difes's refits reach 0.031–0.035 from every start and keep 13 or 14 of 14 oscillators within their width; the ones that move are the ones the report flags (the 400  $\text{cm}^{-1}$  edge, the broad 1127  $\text{cm}^{-1}$  mode). The Bench from Table 6 gives the fitter's result in four visible stages (0.062 → 0.032); from its own suggestions it finds six bands and the two strongest (452, 514  $\text{cm}^{-1}$ ) but not the 577/624 pair nor the weak overtones.

**Verdict.** Achieved where the physics is, in the bands; the disagreement is where the report says its fit was weak. A fourteen-term model is prior knowledge, and the Bench's Import FOCUS model is how it enters.



**Figure 5:** Panels C and D: the oxide film refit from Table 6, and the Bench's guided fit from Table 6 and from its own suggestions.

*Where to look:* Validation against FOCUS, levels C and D; The Bench (manual).

## T11 · Irati's path and FOCUS's exported curves: the same place, to the last digit

**Grade A · exact · class I** (FOCUS's numerical output) · 2026-10-02 · difes 0.14.1 · log L046

**Objective.** Following Irati's five models from the bare metal to the final anisotropic film, does difes land on FOCUS's numbers at every step where FOCUS wrote them down?

**Reference.** Three curves exported from FOCUS by Irati (Azkena-PhysRev\_doitzeko.dat, Azkena-oxido.dat, Azkena-teoriko\_monokristal.dat), 512 points on  $0\text{--}1500\text{ cm}^{-1}$ , seven significant digits; FOCUS's formulas in both project files.

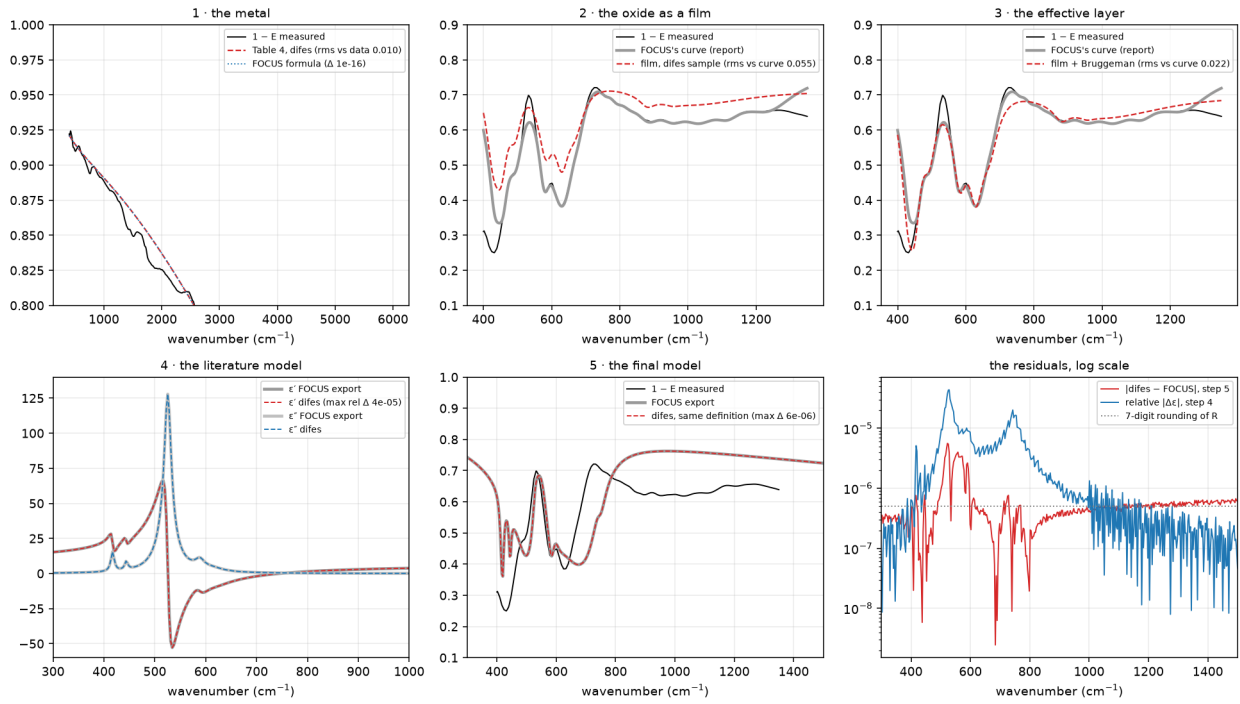
**Data.** The five models of the TFG and of the two exports; the measured  $500\text{ °C}$  emissivities for the against-the-measurement column.

**Physics covered.** Step 1 the metal (Table 4); 2 the oxide as a  $0.8\text{ }\mu\text{m}$  film, eq. 27; 3 a  $0.5\text{ }\mu\text{m}$  Bruggeman mixture of air and oxide over the film; 4 the single-crystal  $\text{Cr}_2\text{O}_3$  dielectric function; 5  $\frac{1}{3}\varepsilon_{\parallel} + \frac{2}{3}\varepsilon_{\perp}$  in a  $0.7\text{ }\mu\text{m}$  film under a  $0.3\text{ }\mu\text{m}$  Bruggeman layer.

**Procedure.** Each step rebuilt as a difes session (native nodes for 1 and 2, FOCUS's own formulas as expression nodes with the project's library for 3 to 5) and evaluated on FOCUS's grid; compared with FOCUS's export where one exists and with FOCUS's formula otherwise.

**Result.** Steps 1 and 2: difes's native optics and FOCUS's formulas agree to  $10^{-16}$ . Step 4: FOCUS's exported  $\varepsilon$  to  $4.4 \times 10^{-5}$  relative on values up to 128. Step 5: FOCUS's exported R to  $5.6 \times 10^{-6}$  at most (rms  $8.6 \times 10^{-7}$ ). Both are the seven-digit rounding of the files. Step 3 is the model of the report's figure 20 (0.022 from the drawn curve).

**Verdict.** Achieved, exactly. Number against number, nothing transcribed, nothing guessed: the final anisotropic model with its effective-medium layer computes in difes what FOCUS wrote.



**Figure 6:** The five steps and the residuals of steps 4 and 5 on a log scale, with the seven-digit rounding of FOCUS's files as the dotted line.

*Where to look:* Irati's path, step by step; difes/examples/irati\_path.py; sessions Irati's path · 1 to 5 in the app.

## T12 · The fluorides of Passerat de Silans et al. (2009): every table computed forward

**Grade B · within the data ·** class II (FOCUS's output as a vector figure) · 2026-10-02 · difes 0.14.2 · log L050

**Objective.** Do the published FOCUS parameters of  $\text{CaF}_2$  and  $\text{BaF}_2$ , from 22 to 500 °C, regenerate the measured reflectivity the paper shows them fitting, with no parameter fitted?

**Reference.** arXiv:0905.4710, Tables I and II (every parameter at six temperatures) and Figs. 1 and 2, vector graphics read exactly.

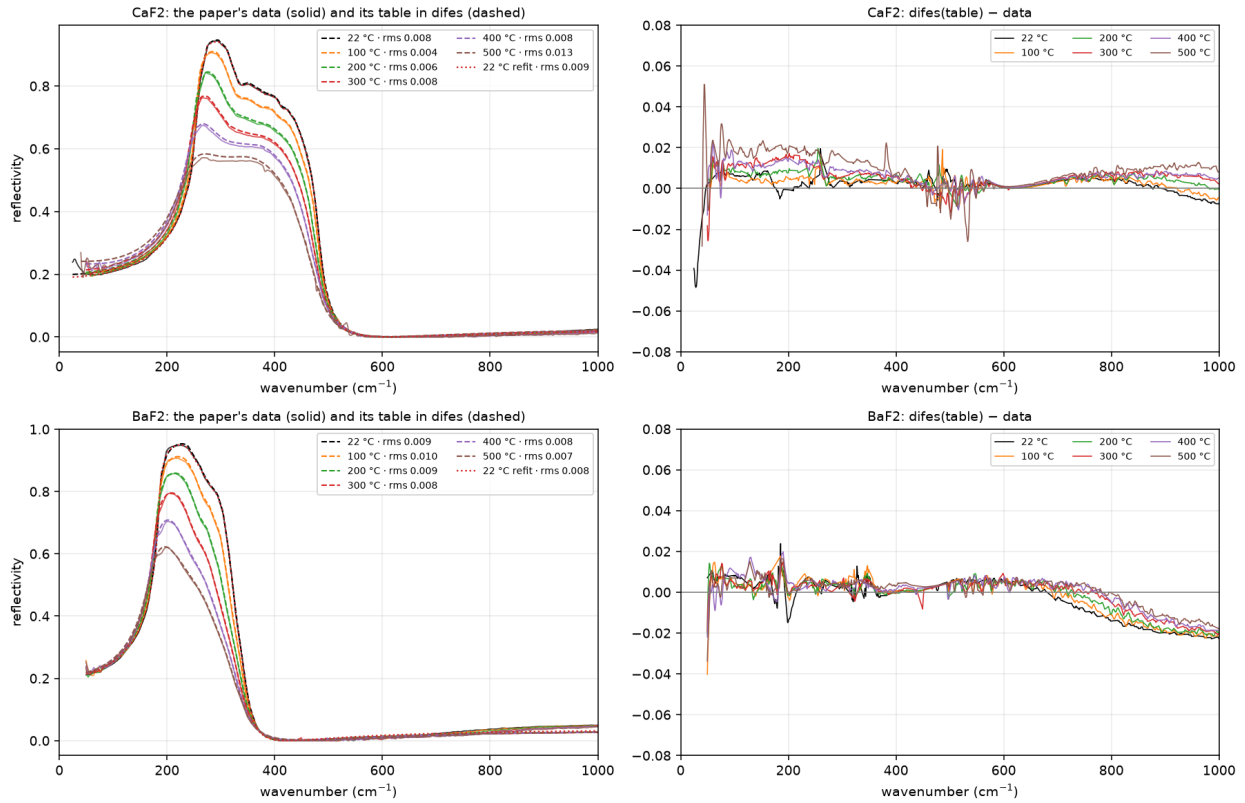
**Data.** Measured reflectivity of  $\text{CaF}_2$  and  $\text{BaF}_2$  at 22, 100, 200, 300, 400 and 500 °C on 0–1000  $\text{cm}^{-1}$ , from the vector paths of the PDF; the paper states a measurement accuracy of 2–3 %.

**Physics covered.** One phonon whose damping is a self-energy of causal Gaussian bands (the paper's eqs. 3–4, FOCUS's anharmonic phonon):  $\text{phonon\_anharmonic}(x, S, \Omega, -2 \Omega \cdot \Sigma \text{ gaussian\_kkg})$  on  $\varepsilon_\infty$ .

**Procedure.** Compute each table forward in difes on the data's grid; rms over 0–1000  $\text{cm}^{-1}$  and over the 100–700  $\text{cm}^{-1}$  band; check passivity; refit at 22 °C with everything free to see whether the printed values are a minimum.

**Result.** rms 0.004–0.013 at every temperature for both materials, every  $\varepsilon$  passive; the refit at 22 °C moves the curve by less than the noise ( $\varepsilon_\infty$  2.02  $\rightarrow$  2.03 for  $\text{CaF}_2$ , 2.12  $\rightarrow$  2.22 for  $\text{BaF}_2$ ).

**Verdict.** Achieved: difes's Gaussian self-energy is FOCUS's, on the material FOCUS's authors chose to show it on, and the printed parameters are a minimum for difes as they were for FOCUS.



**Figure 7:** The paper's measured curves (solid) and its tables computed by difes (dashed) at six temperatures; right, the differences. No parameter fitted.

Where to look: Published works, solved again · 2; difes/examples/papers/fluorides2009/.

## T13 · Sapphire, the same paper: four phonons with Gaussian self-energies

**Grade B** · **within the data** · class II (FOCUS's output as a vector figure) · 2026-10-02 · difes 0.14.3 · log L051

**Objective.** Does the 41-parameter sapphire model of Table III, four phonons each with its own Gaussian self-energy, regenerate the six measured curves of Fig. 3?

**Reference.** The same paper, Table III (all 41 parameters at six temperatures) and Fig. 3, whose six curves are a single vector path cut where the wavenumber jumps back.

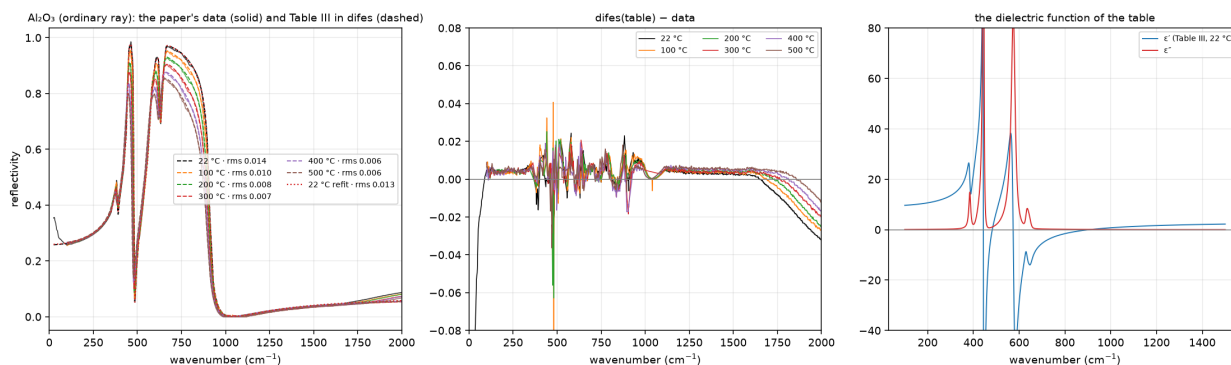
**Data.**  $\text{Al}_2\text{O}_3$ , E<sub>c</sub>, reflectivity on 0–2000  $\text{cm}^{-1}$  at 22 to 500 °C, read exactly from the PDF.

**Physics covered.** Four anharmonic phonons: one Gaussian for phonons 1, 2 and 4, four for phonon 3.

**Procedure.** Forward evaluation at each temperature; rms over 0–2000 and 300–1000  $\text{cm}^{-1}$ ; band maxima model against data; a refit of strengths, frequencies and  $\varepsilon_\infty$  at 22 °C.

**Result.** rms 0.006–0.014 at every temperature, every  $\varepsilon$  passive; band maxima within 0.004 except the sharp 440  $\text{cm}^{-1}$  peak at 22 °C (0.975 against 0.986, the plot's own resolution). The refit moves the frequencies by less than 1 % and the rms from 0.0138 to 0.0131.

**Verdict.** Achieved: the printed values are a minimum for difes too; the one difference is a peak sharper than the figure can draw.



**Figure 8:** The measured reflectivity (solid) and Table III in difes (dashed) at six temperatures; the differences; the dielectric function at 22 °C.

*Where to look:* Published works, solved again · 3.

## T14 · The figure extractors: what a digitised curve is worth

**Grade B** · **within the data** · class IV (physics alone) · 2026-10-02 · difes 0.15.0 · log L052

**Objective.** With what precision do difes’s two extractors read a published figure, so that every digitised comparison in this ledger has a known floor?

**Reference.** Known curves: the extractors are run on figures whose data are also available as files, and on synthetic plots.

**Data.** Bitmap figures at 200 dpi and at the PDF’s native resolution ( $591 \times 395$  pixels for paper 11); vector figures read from PDF paths (4 820 points per curve in paper 17).

**Physics covered.** None; measurement of the method.

**Procedure.** Bitmap: find the frame, calibrate the axes on the ranges or on the tick marks, mask one colour, one point per column for a line or blob centroids for markers, exclude the legend, write an overlay to check by eye. Vector: read the polylines with PyMuPDF and calibrate on the tick-label positions.

**Result.** Measured on known curves drawn as figures (4 October, T39): for a line, rms  $3.9 \times 10^{-3}$  of the axis span at 600 pixels of width,  $2.4 \times 10^{-3}$  at 900 (the pBN paper’s figure),  $1.4 \times 10^{-3}$  at 1500 and  $0.9 \times 10^{-3}$  at 2200 (the TFM’s), with the largest errors on steep edges; for markers  $2.8 \times 10^{-2}$  at 600 pixels and  $1.7 \times 10^{-2}$  at 900, where touching markers merge, falling to  $3.6 \times 10^{-3}$  at 1500. Vector: the precision of the authors’ plotting, effectively exact. Tests pin the extracted point counts of the sample spectra and the precision on a known curve.

**Verdict.** Achieved as a calibrated instrument: a bitmap line can claim a few parts in a thousand of the axis span, markers at low resolution a few percent, a vector figure the file’s digits. The measured values are the digitisation component of the GUM budget (Component.digitised). The extractors are the Digitise panel of the app.

*Where to look:* Digitising figures (manual); difes/src/difes/digitize.py; tests/test\_digitize.py.

## T15 · FOCUS’s sample projects: the spectra read from the binary .pjt files

**Grade A** · **exact** · class I (FOCUS’s numerical output) · 2026-10-02 · difes 0.15.1 · log L055

**Objective.** Can the measured spectra inside FOCUS’s four downloadable sample projects be read from the binary project format, so that the app holds the same data FOCUS does?

**Reference.** The four .pjt files of the FOCUS website (quartz A2, Borofloat glass, calcium phosphate glass,  $\text{SnO}_2$  Raman); later Irati’s exports of the same data as text, which must coincide.

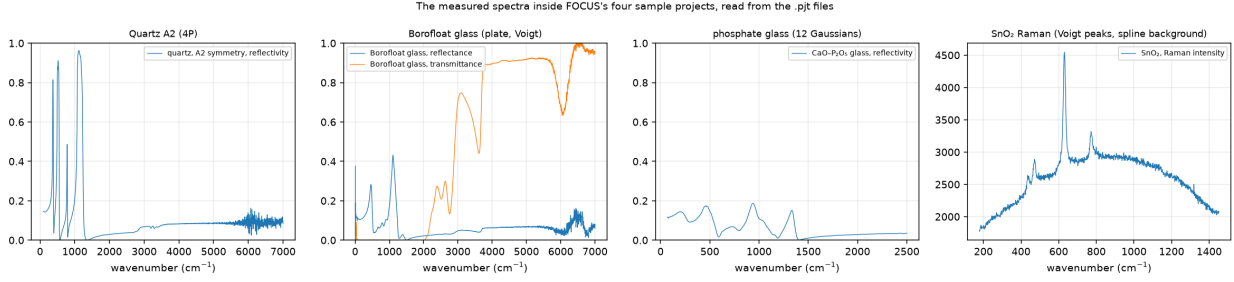
**Data.** Reflectivity of quartz (7 158 points, 99–7 000  $\text{cm}^{-1}$ ), R and T of the Borofloat plate ( $\approx 7\,250$  points each), reflectivity of the phosphate glass (2 523 points, 69–2 500  $\text{cm}^{-1}$ ), Raman intensity of  $\text{SnO}_2$  (1 100 points, 178–1 451  $\text{cm}^{-1}$ ).

**Physics covered.** None; data recovery.

**Procedure.** Scan the file at every byte alignment for runs of little-endian doubles in triples (x, y, 0) with x increasing; keep the runs that read as spectra; write them with the project name and byte offset in the header.

**Result.** All spectra recovered; Irati's later text exports from FOCUS are identical to what was read from the binaries.

**Verdict.** Achieved. The models inside the binaries were not decoded (the parameter record layout is unknown) and came as OSML exports instead (T16 to T19).



**Figure 9:** The measured spectra read from the four .pjt files.

*Where to look:* FOCUS's own sample projects; difes/examples/focus\_samples/extract\_pjt.py; tests/test\_focus\_samples.py.

## T16 · Quartz A2: five four-parameter modes against FOCUS's exported curve

**Grade A · exact · class I** (FOCUS's numerical output) · 2026-10-04 · difes 0.16.0 · log L058

**Objective.** Is difes's four-parameter model FOCUS's, number against number, on FOCUS's own sample project?

**Reference.** FOCUS's IR-Quartz-A2-4P project exported by Irati as OSML with declarations, with FOCUS's computed reflectivity and dielectric function as text (seven significant digits, six for the parameters).

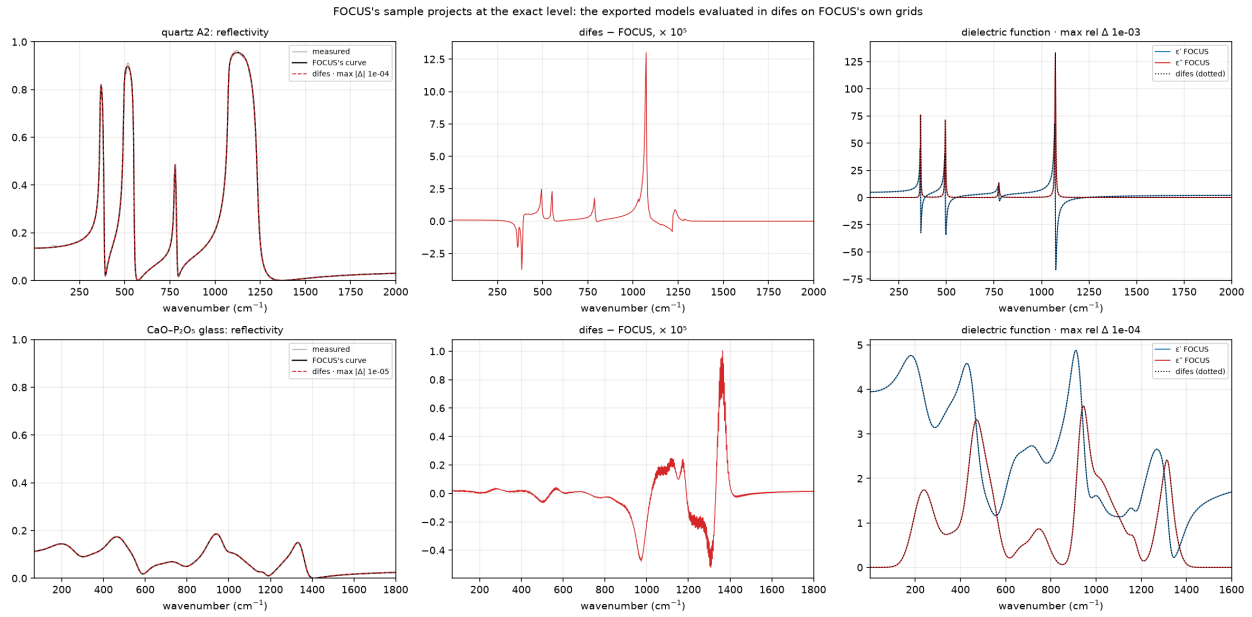
**Data.** FOCUS's curve and the measured reflectivity, 99–7 000  $\text{cm}^{-1}$ ; the exported model:  $\varepsilon_\infty$  2.403 and five four-parameter modes.

**Physics covered.**  $\varepsilon_\infty$  ·  $\Pi$  phonon-4p; bulk reflectivity at normal incidence.

**Procedure.** Evaluate the exported model in difes on FOCUS's grid by two routes (the document's definitions as formulas, difes's native classes) and compare with FOCUS's curve; then refit the measured data from FOCUS's parameters and from a start with every frequency off by 4 % and every damping by 60 %.

**Result.** Reflectivity: median  $8 \times 10^{-7}$ , worst  $1.3 \times 10^{-4}$  on the steepest edge of the 1073  $\text{cm}^{-1}$  mode, where the six digits of the exported parameters move the edge by the amount seen;  $\varepsilon$  to  $10^{-3}$  relative at the poles. From FOCUS's parameters nothing moves (rms 0.0046, the data's scatter); from the wrong start FOCUS's values come back, frequencies within 0.01 % in the median.

**Verdict.** Achieved, exactly. The four-parameter question is closed: the gap between printed tables and drawn fits in the literature (T23 to T26) is outside the model.



**Figure 10:** Quartz A2 and the phosphate glass: the data, FOCUS's exported curve and difes's evaluation (dashed); the difference times  $10^5$ ; the dielectric function.

Where to look: FOCUS's own sample projects · the exact level; difes/examples/focus\_samples/exact\_tests.py; tests/test\_focus\_exact.py.

## T17 · The phosphate glass: twelve causal Gaussians against FOCUS's export

**Grade A** · **exact** · class I (FOCUS's numerical output) · 2026-10-04 · difes 0.16.0 · log L058

**Objective.** Is difes's causal Gaussian band FOCUS's gaussian-kkg, on a disordered solid where it is the natural profile?

**Reference.** FOCUS's IRglass-CaO-PO4-Gaussian project exported as OSML, with FOCUS's computed reflectivity and  $\varepsilon$ .

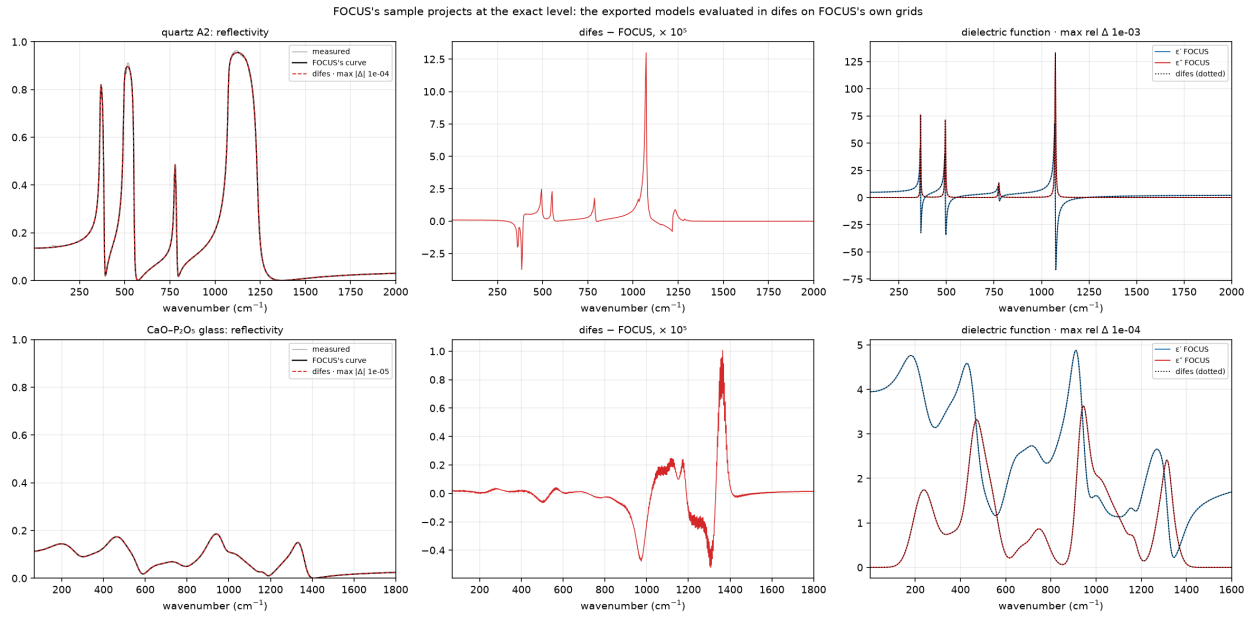
**Data.** Reflectivity 69–2 500  $\text{cm}^{-1}$ , 2 523 points; the exported model:  $\varepsilon_\infty$  2.21 and twelve gaussian-kkg bands.

**Physics covered.** Sum of causal Gaussian bands (Dawson pair) on  $\varepsilon_\infty$ ; bulk reflectivity.

**Procedure.** As for quartz: two evaluation routes against FOCUS's curve; refits from FOCUS's parameters and from the 4 %/60 % perturbed start.

**Result.** Reflectivity to  $10^{-5}$  (median  $2 \times 10^{-7}$ );  $\varepsilon$  to  $10^{-4}$  relative. Both starts return FOCUS's values, frequencies within 0.05 %.

**Verdict.** Achieved, exactly; the fitter recovers FOCUS's twelve bands from a wrong start.



**Figure 11:** Right half: the phosphate glass, FOCUS's exported curve and difes's; the difference times  $10^5$ ; the dielectric function.

*Where to look:* FOCUS's own sample projects · the exact level.

## T18 · The Borofloat plate: 32 causal Voigt bands, R and T of a 1 mm slab

**Grade C · explained · class I** (FOCUS's numerical output) · 2026-10-04 · difes 0.16.0 · log L058, L059

**Objective.** Does difes compute FOCUS's reflectance and transmittance of a glass plate from FOCUS's 32-band Voigt model, and its dielectric function?

**Reference.** FOCUS's IRGlass-Borofloat-Voigt project exported as OSML, with FOCUS's computed R, T and  $\varepsilon$ ; FOCUS's library formulas Reflectance-E and Transmittance-E.

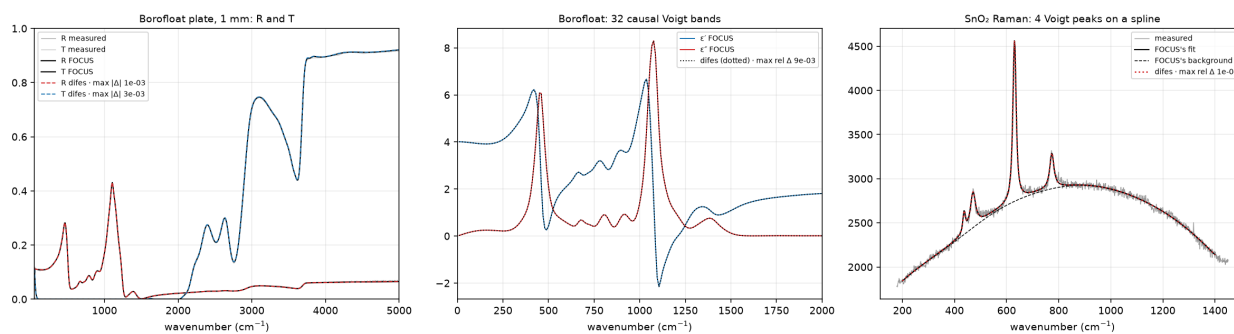
**Data.** Measured R and T of the plate ( $\approx 7\,250$  points each); FOCUS's curves on the same grid; the model: 32 voigt-kkg bands, thickness 1 mm.

**Physics covered.** Causal Voigt bands (Faddeeva function with Kramers-Kronig partner); incoherent plate R and T.

**Procedure.** Implement voigt-kkg from the Faddeeva function; evaluate the exported model; compare  $\varepsilon$ , bulk R, plate R and T with FOCUS's; compare difes's slab optics with FOCUS's library formulas; test three explanations of the residual (Humlíček's approximation, parameter rounding, pseudo-Voigt forms).

**Result.** One band, the one at  $81\text{ cm}^{-1}$  wider than its centre, was first low by a factor 2.2: FOCUS normalises the mirrored pair so that  $\varepsilon''$  at the band centre equals the amplitude, a convention invisible for narrow bands. With it  $\varepsilon$  agrees to 0.9 %; plate R to  $1.5 \times 10^{-3}$ , T to  $2.8 \times 10^{-3}$ ; difes's slab optics reproduce Reflectance-E and Transmittance-E to the same figures. The 0.9 % is not Humlíček ( $1.4 \times 10^{-4}$ ), not rounding, not a pseudo-Voigt.

**Verdict.** Explained. The convention is adopted inside difes's voigt\_kkg so that FOCUS files read faithfully; the residual is FOCUS's own approximation of the Voigt function, pinned down by the one-band exports (T20). Its effect on R and T is below any measurement.



**Figure 12:** The Borofloat plate's measured R and T, FOCUS's curves and difes's (dashed); the 32-band dielectric function; right, the SnO<sub>2</sub> Raman fit.

Where to look: FOCUS's own sample projects · FOCUS's causal Voigt, decoded; difes/examples/focus\_samples/exact\_borofloat\_residual\_diagnostics.py.

## T19 · SnO<sub>2</sub> Raman: four Voigt peaks on a cubic-spline base line

**Grade A** · **exact** · class I (FOCUS's numerical output) · 2026-10-04 · difes 0.16.0 · log L058

**Objective.** Outside the optical scope, does difes's language reproduce FOCUS's peaks-plus-base-line sum and FOCUS's spline exactly?

**Reference.** FOCUS's Raman-SnO<sub>2</sub> project exported as OSML, with FOCUS's fitted sum and its base line as text.

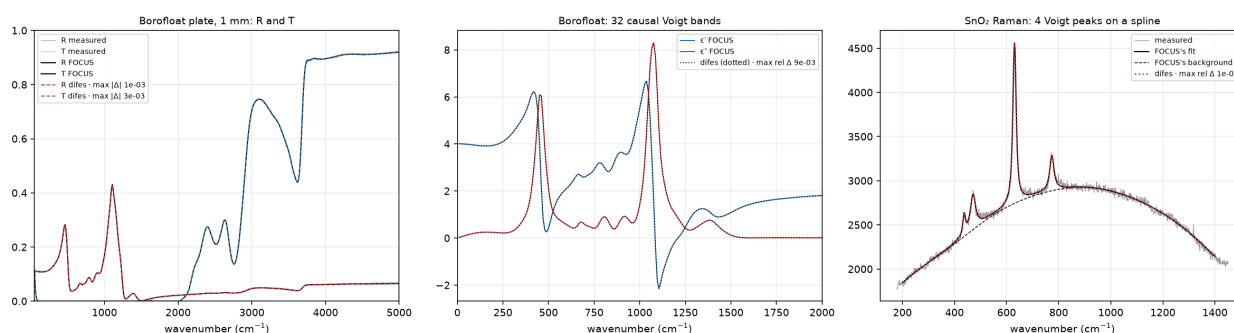
**Data.** Raman intensity 178–1 451 cm<sup>-1</sup>, 1 100 points; four Voigt peaks; a cubic spline through nine points.

**Physics covered.** None of the dielectric kind: Voigt peak shapes and a spline.

**Procedure.** Implement cubic\_spline in the math language; test the four end conditions against FOCUS's base line; evaluate peaks plus spline against FOCUS's sum.

**Result.** The natural spline matches to  $4.6 \times 10^{-6}$  (the other end conditions fail at  $10^{-3}$ ); peaks plus spline against FOCUS's sum to  $1.2 \times 10^{-4}$ .

**Verdict.** Achieved, exactly. FOCUS's base line is a natural cubic spline, and the fitter and profiles work outside reflectivity.



**Figure 13:** Right panel: the Raman spectrum, FOCUS's fit and background, difes's sum (dotted).

Where to look: FOCUS's own sample projects.

## T20 · Four one-band exports: the Gaussian convention closed, FOCUS's Voigt characterised

**Grade C** · **explained** · class I (FOCUS's numerical output) · 2026-10-04 · difes 0.16.0 · log L061

**Objective.** Band by band, what exactly do FOCUS's gaussian-kkg and voigt-kkg compute, and does difes compute the same?

**Reference.** Four one-band projects built and exported by Irati on request: voigt-kkg(x, 1, 1000, 100, w\_L) for w\_L = 0.1, 30 and 200, and gaussian-kkg(x, 1, 100, 400), each as FOCUS's  $\varepsilon$  on 0–3000 cm<sup>-1</sup>.

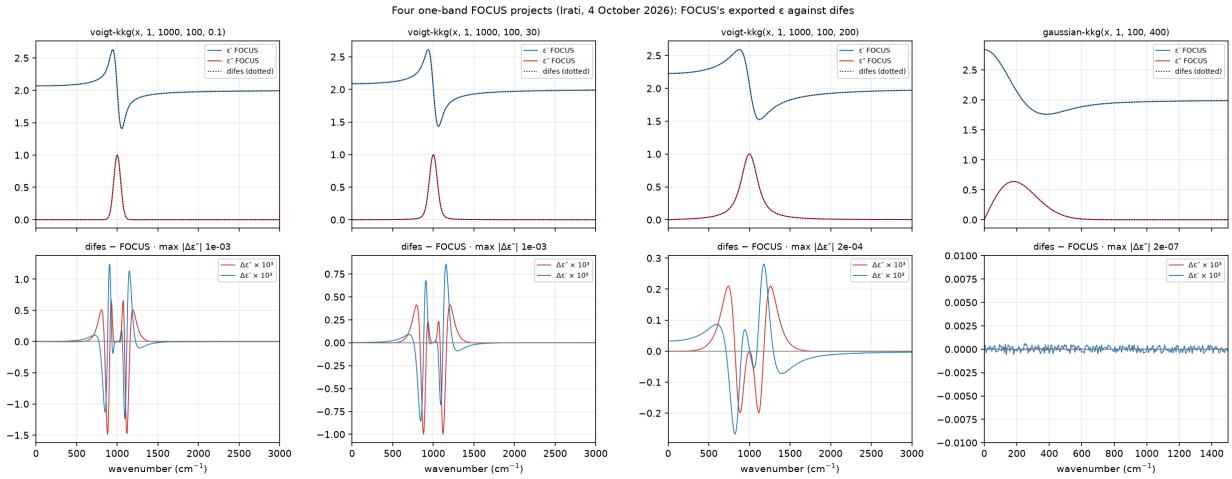
**Data.** FOCUS's  $\varepsilon'$  and  $\varepsilon''$  of each band, seven digits; no measurement.

**Physics covered.** The causal Gaussian pair; the causal Voigt from the Gaussian limit to a Lorentzian width twice the Gaussian one; a Gaussian four times wider than its centre, where any renormalisation would show.

**Procedure.** Evaluate difes's two profiles on FOCUS's grid; compare peak, FWHM and area; compare difes's exact Faddeeva evaluation and the standard approximations (Humlíček w4, Weideman at three orders, Gauss–Hermite at three orders, numerical convolution) with FOCUS's residual.

**Result.** Gaussian:  $\varepsilon'$  and  $\varepsilon''$  to  $6 \times 10^{-7}$  with the published equation; a centre-normalised pair would be off by a factor 2. Voigt: peak, FWHM and area identical to four digits; the residual is an oscillation of  $1.5 \times 10^{-3}$  of the peak at w\_L 0.1,  $1.0 \times 10^{-3}$  at 30,  $2 \times 10^{-4}$  at 200, reproduced by none of the standard algorithms.

**Verdict.** The Gaussian is achieved exactly and the question whether FOCUS renormalises it is closed (it does not). The Voigt is explained: difes evaluates the exact function, FOCUS its own approximation good to a part in a thousand of the band; the per-band size accounts for the whole Borofloat 0.9 %. difes does not imitate the approximation.



**Figure 14:** FOCUS's  $\varepsilon'$  and  $\varepsilon''$  for each one-band project with difes over them; below, the differences times  $10^3$ .

Where to look: FOCUS's own sample projects · four one-band projects; Causal Voigt (theory); difes/examples/focus\_samples/single\_bands.py.

## T21 · Pyrolytic boron nitride, González de Arrieta et al. (2017): the 691 °C fit

**Grade B · within the data** · class III (printed tables and bitmap figures) · 2026-10-02 · difes 0.14.2 · log L050

**Objective.** Fitting the digitised 691 °C emissivity of the paper with the paper's model, does difes reach the curve FOCUS drew on the same data?

**Reference.** J. Quant. Spectrosc. Radiat. Transfer 194 (2017) 1–6, Fig. 3: the 691 °C data with FOCUS's fit drawn on it; the paper's model.

**Data.** Emissivity 4–16  $\mu\text{m}$  at 691 °C, digitised from the bitmap figure (markers and the fitted line separately); precision about  $5 \times 10^{-3}$ .

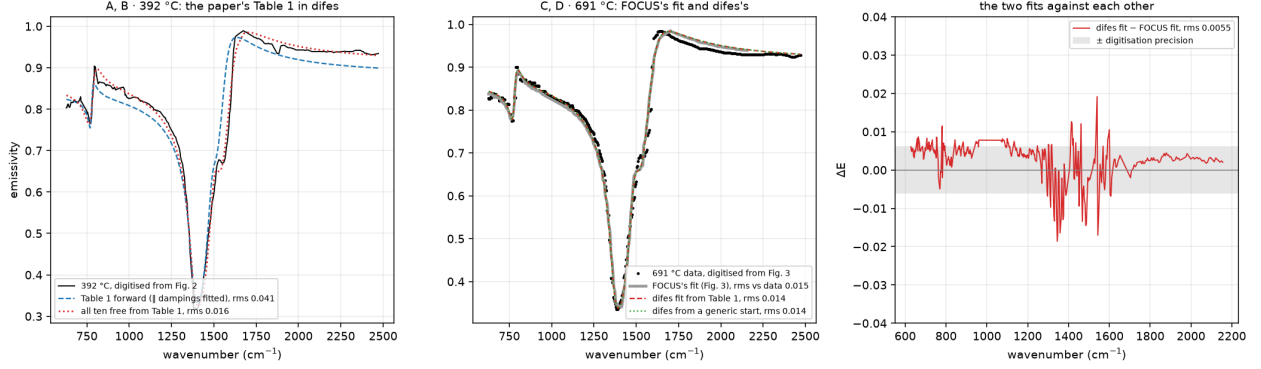
**Physics covered.** Two Kurosawa modes ( and to the c axis), the random-orientation average  $\frac{1}{3} \varepsilon_{\parallel} + \frac{2}{3} \varepsilon_{\perp}$ , a Maxwell–Garnett correction for carbon inclusions ( $\varepsilon = 5$ ,  $f = 8\%$ ); emissivity of an opaque sample.

**Procedure.** Fit all parameters from Table 1 and from a generic start (frequencies  $-5\%$ , dampings 50); compare with the digitised FOCUS curve and with the points.

**Result.** rms 0.014 against the points, where FOCUS's own curve sits 0.015 from the same points; difes's curve and FOCUS's agree to rms 0.0055, the digitisation precision; the generic start gives the same curve.

**Verdict.** Achieved within the digitisation: both programs draw the same fit through the same points with the same model.

**Open point.** Digitisation uncertainty, measured (T39):  $2.4 \times 10^{-3}$  of the axis span per curve at this figure's 900 pixels, so  $3.4 \times 10^{-3}$  for the difference of two digitised curves; the rms of 0.0055 between difes's and FOCUS's fits is 1.6 times that, within the expanded uncertainty at  $k = 2$ . The agreement cannot be stated more finely from this figure.



**Figure 15:** Middle: the 691 °C data of Fig. 3 with FOCUS's fitted curve (grey) and difes's fits; right: difes's fit minus FOCUS's against the digitisation precision.

Where to look: Published works, solved again · 1; difes/examples/papers/pbn2017/reproduce\_pbn.py.

## T22 · Boron nitride, the printed table and the thesis figure: one number in question

**Grade D · open ·** class III (printed tables and bitmap figures) · 2026-10-04 · difes 0.16.0 · log L050, L062

**Objective.** Does the paper's Table 1 (and the TFM's Table 4.2 behind it) regenerate the emissivity and the dielectric function the figures draw?

**Reference.** The paper's Table 1 and Fig. 2 (392 °C); the first author's TFM (UPV/EHU, 2016): Table 4.2 at eight temperatures and Fig. 4.11, the model's own  $\epsilon'$  and  $\epsilon''$  at 691 °C drawn by the fitting program, digitised with the axes calibrated on the tick marks.

**Data.** The 392 °C spectrum of Fig. 2; the  $\epsilon'$  and  $\epsilon''$  curves of Fig. 4.11 on 4–16  $\mu\text{m}$ ; both bitmaps.

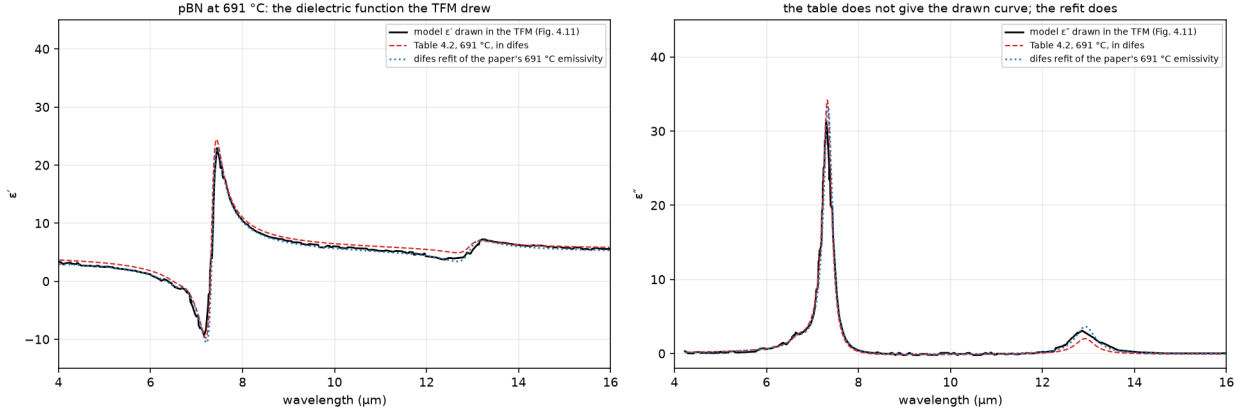
**Physics covered.** As T21; the high-frequency level of  $\epsilon'$  is set by  $\epsilon_\infty$ .

**Procedure.** Table 1 forward on the 392 °C spectrum (the two unprinted parallel dampings fitted); refit with all ten parameters free; Table 4.2's 691 °C column evaluated in difes against the drawn  $\epsilon$ ; difes's refit of the paper's 691 °C data against the same curves; the  $\epsilon_\infty$  that reproduces the drawn level with the table's other values.

**Result.** Table 1 gives  $E = 0.90$  above 2300  $\text{cm}^{-1}$  where the spectrum reads 0.94 and the band edge 50  $\text{cm}^{-1}$  too low (rms 0.041); the refit reaches 0.016 with  $\epsilon_\infty$  5.19  $\rightarrow$  3.89. In the TFM the drawn model has  $\epsilon' = 2.76$  at 4–5  $\mu\text{m}$  and a maximum of 22.9; Table 4.2 in difes gives 3.39 and 24.4; difes's refit gives 2.70 and 22.7. The  $\epsilon_\infty$  that reproduces the drawn level is 3.89, against 5.18 printed and 4.10 refitted; the TO frequencies agree in all three.

**Verdict.** Open, and narrowed to one number: the printed  $\epsilon_\infty$  is not the value of the plotted model, in the thesis as in the paper. The question is put to the author as the brief's second request; the model algebra is not in doubt (T16, T21). Digitisation (T39): at the TFM figure's 2200 pixels the line precision is  $0.9 \times 10^{-3}$  of the span, 0.05 on the  $\epsilon'$  axis of 50; the differences at issue (0.6 and 0.9 rms, 0.63 at the high-frequency level) are ten to twenty times larger and are not digitisation.

**Open point.** Is the printed  $\varepsilon_\infty$  the value of the plotted model, or a different quantity or convention? The author's FOCUS project answers it in a minute.



**Figure 16:** The model  $\varepsilon'$  and  $\varepsilon''$  drawn in the TFM at 691 °C (black), Table 4.2 evaluated in difes (dashed), difes's refit of the paper's 691 °C emissivity (dotted).

*Where to look:* Published works, solved again · 1, the TFM; difes/examples/papers/pbn2017/tfm\_inigo.py.

## T23 · The literature's printed four-parameter tables: 36 of them, evaluated

**Grade A** · **exact** · class IV (physics alone) · 2026-10-02 · difes 0.15.1 · log L056

**Objective.** Evaluating every four-parameter table the FOCUS literature prints, does difes's model algebra reproduce an independent arithmetic, and what do the tables themselves imply?

**Reference.** The audit of 18 FOCUS papers (68 datasets, 797 numbers) and its independent calculation of  $\varepsilon(0) = \varepsilon_\infty \Pi (\Omega_{\text{LO}}/\Omega_{\text{TO}})^2$  for each table.

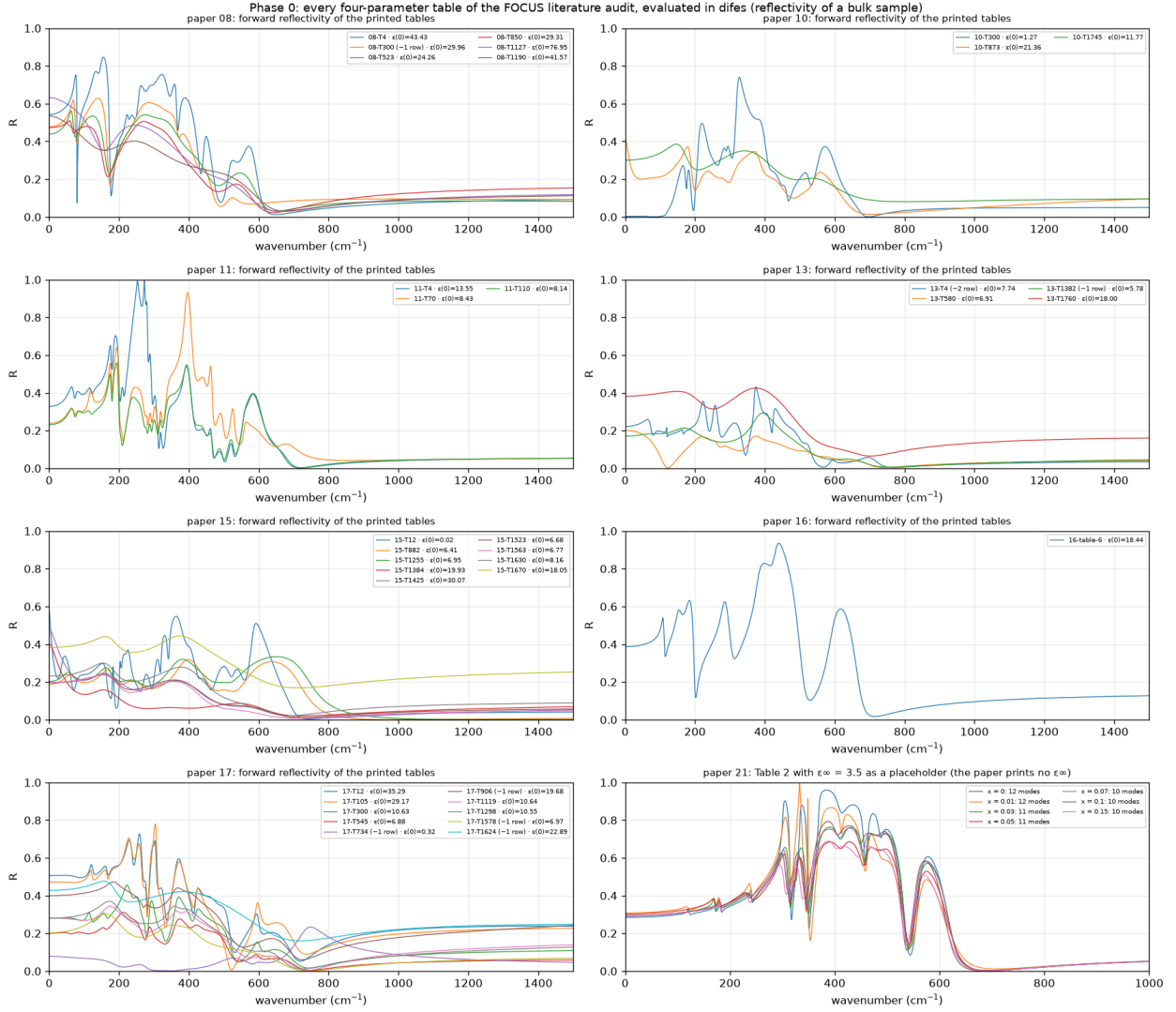
**Data.** 36 four-parameter tables from papers 08, 10, 11, 13, 15, 16, 17, plus the seven compositions of paper 21 with a placeholder  $\varepsilon_\infty$ ; malformed cells left out and counted.

**Physics covered.** Factorised four-parameter modes; the static limit; passivity of  $\varepsilon''$  on 1–5000  $\text{cm}^{-1}$ ; forward bulk reflectivity.

**Procedure.** Load the audit's parameter file directly into difes models; compute  $\varepsilon(0)$ , passivity and R for each; compare  $\varepsilon(0)$  with the audit's arithmetic; reproduce the known-bad rows.

**Result.**  $\varepsilon(0)$  agrees to better than  $10^{-9}$  relative in 25 of 25 complete tables (the 7 others differ exactly by the row left out). Only 14 of 43 tables are passive as printed: the four-parameter product is not passive by construction, and paper 16's seven-mode table has  $\varepsilon'' < 0$  from 50 to 253  $\text{cm}^{-1}$ . The known-bad rows reproduce as flagged (15-T12 gives  $\varepsilon(0) = 0.018$ , 10-T300 1.27, 17-T734 0.32).

**Verdict.** Achieved for the algebra, over 36 independent parameter sets. What it does not establish is agreement with FOCUS, since no FOCUS curve is involved; that came with T16.



**Figure 17:** The forward reflectivity of every printed table, paper by paper.

Where to look: The FOCUS literature, reproduced · Phase 0; Literature search and audit; difes/examples/literature/audit\_rphase0\_tables.py.

**T24 · Paper 17, hexagonal  $\text{ErMnO}_3$ : exact vector data, a table that does not regenerate its fit**

**Grade D · open · class II** (FOCUS's output as a vector figure) · 2026-10-02 · difes 0.15.1 · log L056

**Objective.** On exact data, do the supplement's four-parameter tables at 300 K and 12 K regenerate the fits the supplement shows on the data, and can difes fit the data from them?

**Reference.** Massa et al., Phys. Rev. B 102, 134305 (2020): Fig. 2a as a vector drawing (4 820 points per curve); Tables SII and SIV of the supplement; Figs. S4 and S6 showing those fits on the data.

**Data.** Near-normal reflectivity at 300 K and 12 K, read exactly from the PDF.

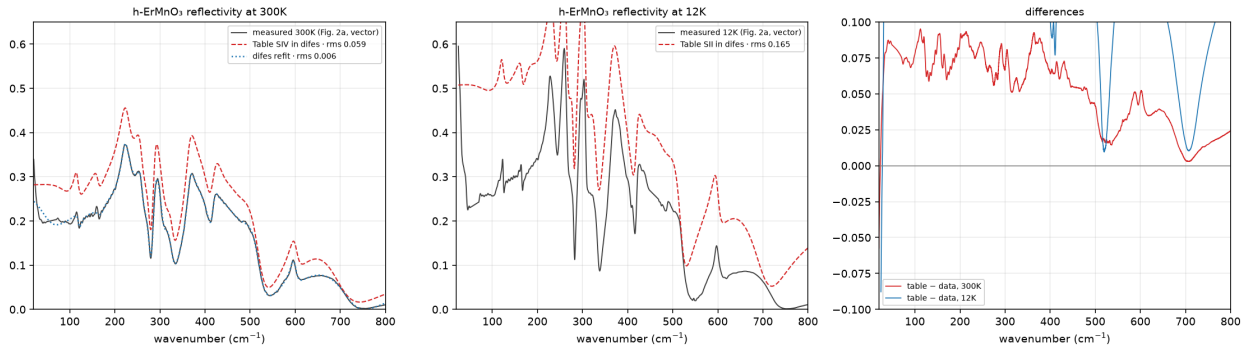
**Physics covered.** Sixteen (300 K) and eighteen (12 K) four-parameter modes with  $\varepsilon_\infty$  1.20, a broad mid-infrared row included.

**Procedure.** Table SIV forward on the 300 K data; the lever analysed (the broad row TO 3852, LO 7446 multiplies the far infrared by 3.7); refit from the table with all 65 parameters free; Table SII forward at 12 K.

**Result.** Table SIV sits 0.06–0.09 above the data through the phonon region (rms 0.059; its  $\varepsilon(0) = 10.6$  gives  $R(50) = 0.28$  where the data read 0.20): the printed table does not regenerate Fig. S6. difes’s refit reaches rms 0.0065 on 4 820 points, 12 of 15 phonon frequencies moving less than 4 %. Table SII at 12 K: rms 0.165, the same offset.

**Verdict.** Open: difes fits the data at the data’s precision, but the published table is not the published fit. A fact about the publication, to be settled by the authors’ export (and by the scale hypothesis, T26).

**Open point.** The authors’ FOCUS project or OSML export with the fitted curve.



**Figure 18:** 300 K and 12 K: the vector data, the printed tables in difes (dashed), difes’s refit (dotted); right, table minus data.

*Where to look:* The FOCUS literature, reproduced · Paper 17.

## T25 · Papers 16 and 11: $\text{Nd}(\text{Mg}_{0.5}\text{Sn}_{0.49})\text{O}_3$ and $\text{NdMnO}_3$ from bitmap figures

**Grade D · open · class III** (printed tables and bitmap figures) · 2026-10-03 · difes 0.15.1 · log L056, L057

**Objective.** Do the printed tables of two more FOCUS papers regenerate their drawn fits, and does difes fit the digitised data?

**Reference.** Paper 16 (Applied Physics A, 2020, preprint): Table 6, seven modes,  $\varepsilon_\infty$  5.01,  $\varepsilon(0)$  18.44; Fig. 9a at 200 dpi. Paper 11 (Massa et al., J. Phys.: Condens. Matter 25, 395601, 2013): Table 1 at 4 K, 25 modes,  $\varepsilon_\infty$  2.74; Fig. 2 on a log axis at its native  $591 \times 395$  pixels.

**Data.** Reflectivity of the ceramic (hollow markers) and FOCUS’s drawn fit; the 4 K  $\text{NdMnO}_3$  spectrum with its fit, 60–2000  $\text{cm}^{-1}$ .

**Physics covered.** Factorised four-parameter modes, seven and twenty-five.

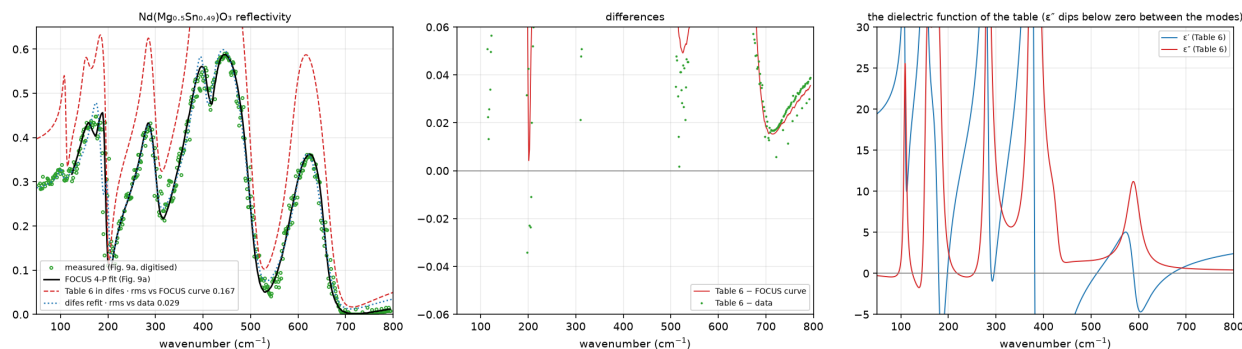
**Procedure.** Tables forward against the drawn fits; refits from the tables; a damping-only refit for paper 16.

**Result.** Paper 16: the table gives peaks of 0.93 where the figure has 0.59 (rms 0.167); self-consistent with  $\varepsilon(0)$  18.44 but not with the figure; difes’s refit reaches 0.029, the digitisation level of hollow markers (FOCUS’s own curve is 0.034 from them), with  $\varepsilon_\infty$  5.01  $\rightarrow$  3.76. Paper 11: rms 0.166 as printed, agreement above 1000  $\text{cm}^{-1}$  (so  $\varepsilon_\infty$  is not the issue); difes’s refit of 101 parameters reaches 0.0125, 15 of 25 frequencies within 5 % of the table.

**Verdict.** Open: the same pattern as paper 17, in the same direction, from two groups. Questions for the authors, not results about difes. Digitisation (T39): paper 16’s figure is 550 pixels wide and its data are hollow markers, for which the measured precision is  $2.8 \times 10^{-2}$  of the span, 0.02 on this axis; difes’s refit rms of 0.029 against those markers is at that limit, and FOCUS’s own curve sits 0.034 from them. Paper 11’s

figure is 591 pixels wide:  $3.9 \times 10^{-3}$  of a 0.6 span is  $2.3 \times 10^{-3}$  per curve, far below the 0.166 between the printed table and the drawn fit.

**Open point.** The authors' exports; and whether a factor was applied to the model reflectivity (T26).



**Figure 19:** Paper 16: the digitised data and FOCUS curve, Table 6 in difes (dashed), difes's refit (dotted); the table's dielectric function with  $\varepsilon''$  below zero between 50 and 253  $\text{cm}^{-1}$ .

*Where to look:* The FOCUS literature, reproduced · Papers 16 and 11.

## T26 · The scale hypothesis: one unprinted factor on the published reflectivity

**Grade D** · **open** · class III (printed tables and bitmap figures) · 2026-10-03 · difes 0.15.1 · log L057

**Objective.** Can one multiplicative factor on the model reflectivity, the usual way to fit a porous ceramic, explain why three printed tables sit above their drawn fits?

**Reference.** The drawn fits of papers 16, 17 and 11; FOCUS's own Factor object checked as a possible source.

**Data.** The digitised and vector curves of T24 and T25.

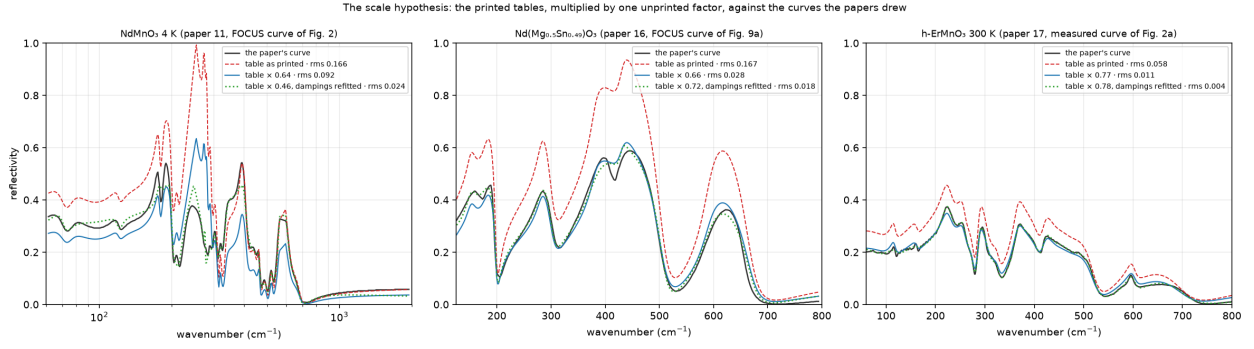
**Physics covered.**  $R_{\text{fit}} = s \cdot R_{\text{model}}$  with  $s$  a constant; dampings refitted with frequencies and  $\varepsilon_{\infty}$  fixed as a second step.

**Procedure.** Multiply each table's reflectivity by one  $s$  fitted to the drawn curve; then refit  $s$  and the dampings only.

**Result.** Paper 16:  $s = 0.66$ , rms  $0.167 \rightarrow 0.028$  (the digitisation level). Paper 17:  $s = 0.77$ , rms  $0.058 \rightarrow 0.011$ . Paper 11:  $s = 0.64$ ,  $0.166 \rightarrow 0.092$ ; with dampings refitted  $s = 0.46$ , rms  $0.024$ : the factor explains half, the 4 K table has other problems (the audit's issue N02). FOCUS's Factor is only the name of its product container.

**Verdict.** Open: a hypothesis that fits two of three cases to the digitisation level, consistent with the boron nitride direction (an emissivity below the figure is what a scaled  $R$  turned into  $1 - R$  gives). It is a hypothesis until an author confirms it; what is certain is that the tables' frequencies are right and the algebra is not in question.

**Open point.** One direct question to the authors of 11, 16 and 17: was a multiplicative factor applied to the model reflectivity, and which?



**Figure 20:** The paper's curve (grey), the table as printed (dashed), the table times one factor (solid), and with the dampings refitted (dotted), for the three cases.

Where to look: The FOCUS literature, reproduced · The scale hypothesis; difes/examples/literature/scale\_hypothesis.py.

## T27 · Iñigo's thesis (2020): four vanadium alloys, Drude, from a generic start

**Grade A** · **exact** · class II (FOCUS's output as a vector figure) · 2026-10-04 · difes 0.16.0 · log L063

**Objective.** Given the same Drude model and the same measured points, does difes land on the four curves FOCUS fitted in the author's doctoral thesis?

**Reference.** The thesis's Fig. 5.9 (UPV/EHU, 2020), a vector figure: the normal spectral emissivity of four V-4Cr-4Ti alloys (EBM, EBM-HT, MA, MA-HT) at 973 K with FOCUS's Drude fits (eq. 2.34); Fig. 5.10 shows  $\sigma = \omega_p^2/(4\gamma)$  and  $\varepsilon_\infty$  as charts, no table.

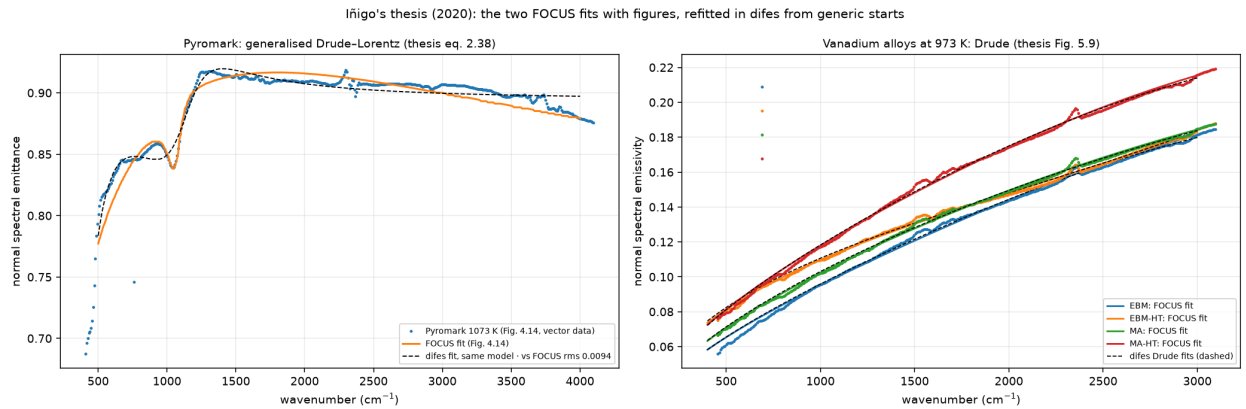
**Data.** About 340 points per alloy on 400–3100 cm<sup>-1</sup>, and FOCUS's four fitted curves, read exactly from the PDF paths.

**Physics covered.**  $\varepsilon_\infty - \omega_p^2/(x^2 + i x)$ ; emissivity  $1 - R$  of an opaque metal.

**Procedure.** Fit the Drude model to each alloy's points from a generic start; compare difes's curve with FOCUS's drawn curve; derive  $\sigma$  and compare with the range of Fig. 5.10.

**Result.** difes's curve against FOCUS's: rms 0.0007 (EBM), 0.0008 (EBM-HT), 0.0005 (MA), 0.0006 (MA-HT), maxima 0.0007–0.0015, the line width of the figure;  $\sigma$  of 76 800, 39 100, 63 600 and 49 700 cm<sup>-1</sup>, inside the 0–80 000 axis of Fig. 5.10. The  $\varepsilon_\infty$  values are large and poorly determined, as a Drude  $\varepsilon_\infty$  is when only emissivity below 3100 cm<sup>-1</sup> constrains it.

**Verdict.** Achieved, exactly at the figure's precision: the FOCUS expert's own fits, met from scratch.



**Figure 21:** Right: the four alloys at 973 K from Fig. 5.9, FOCUS's Drude curves in colour and difes's fits dashed on top. Left: the Pyromark case (T28).

Where to look: Published works, solved again · the PhD thesis; difes/examples/papers/inigo\_thesis/thesis\_fits.py.

## T28 · Iñigo's thesis: the Pyromark coating, two minima of one model

**Grade C** · **explained** · class II (FOCUS's output as a vector figure) · 2026-10-04 · difes 0.16.0 · log L063

**Objective.** With the thesis's generalised Drude–Lorentz model and the measured emittance of a Pyromark coating at 1073 K, does difes reproduce FOCUS's fit, called semi-quantitative in the text?

**Reference.** The thesis's Fig. 4.14 (vector): the measured normal spectral emittance and FOCUS's fit with the model of eq. 2.38.

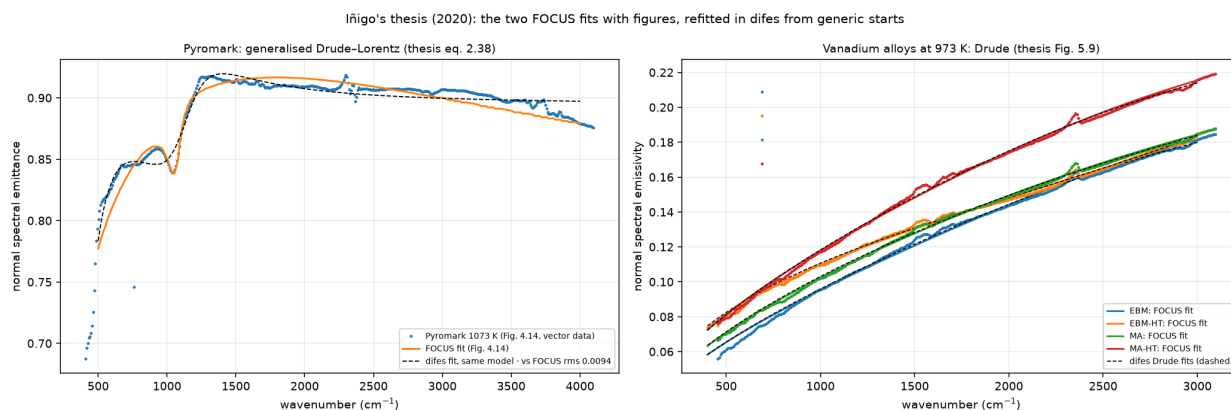
**Data.** 480 points on 409–4097  $\text{cm}^{-1}$  and FOCUS's fitted curve (213 points), read exactly from the PDF.

**Physics covered.**  $\varepsilon_\infty$  · (phonon-4p – extended Drude with two dampings): eight parameters, one smooth spectrum.

**Procedure.** Fit the model to the data from generic starts; compare with FOCUS's curve; then fit the same model to FOCUS's drawn curve itself to see whether it is a member of the family.

**Result.** difes's fit: rms 0.0086 against the points; FOCUS's drawn curve: 0.0130 against the same points; the two curves 0.0094 rms apart, 0.022 at most. The same model fitted to FOCUS's curve reproduces it to 0.0004 with a narrow oscillator at 1060  $\text{cm}^{-1}$  and a heavily overdamped Drude term, against a wide oscillator at 1083  $\text{cm}^{-1}$  and a narrow Drude term in difes's solution.

**Verdict.** Explained: both results are members of the model family and the optics agree; the two programs stopped in different minima of a shallow eight-parameter problem, difes in the lower one. A statement about the problem, not about either program.



**Figure 22:** Left: Pyromark at 1073 K, the measured points and FOCUS's fit from Fig. 4.14, and difes's fit of the same model (dashed).

*Where to look:* Published works, solved again · the PhD thesis.

## T29 · Papers that cannot be reproduced yet

**Grade E** · **blocked** · class III (printed tables and bitmap figures) · 2026-10-04 · difes 0.16.0 · log L056

**Objective.** Which of the 18 FOCUS papers held cannot be tested, and why?

**Reference.** The audit's inventory of each paper's printed material.

**Data.** Paper 03 (silica, eight causal Gaussians fitted to R built from Popova's  $n$ ,  $k$ ): table complete, Popova's table needed. 04 (the CFRC method paper): coefficients, no synthetic inputs. 05 (La–Nd stannate): only aggregates printed. 12 (gold nanoparticles): KK of Palik's data, table needed. 14 (graphene on SiC): the model is in a missing supplement. 15 (o/c-ErMnO<sub>3</sub>): blocked by a printed LO of 14.6 at 12 K. 21 ((Y, La)<sub>2</sub>O<sub>3</sub>): no  $\varepsilon_\infty$  printed, to be fitted against the bitmap Fig. 6. 08, 10, 13: Phase 0 done, figure panels too small or tables malformed for a curve comparison. 01, 09, 18, 19: Raman decompositions, outside the optical scope.

**Physics covered.** Causal Gaussians, CFRC, Kramers–Kronig of tabulated data, four-parameter products.

**Procedure.** None run beyond Phase 0 (T23).

**Result.** Blocked by external data or by the publications themselves.

**Verdict.** Not achieved, by lack of inputs rather than of means: each becomes an exact comparison the day an author's export or the external table arrives.

**Open point.** Popova's n, k table; the reference model of the CFRC quartz example; the authors' oscillator exports.

*Where to look:* The FOCUS literature, reproduced · The ledger.

### T30 · What difes does not have yet

**Grade E · blocked ·** class I (FOCUS's numerical output) · 2026-10-04 · difes 0.16.0 · log L060

**Objective.** Which FOCUS capabilities have no difes counterpart to test?

**Reference.** FOCUS's help and Irati's projects.

**Data.** The profiles lorentzian-kkg, PearsonVII and the area-normalised peaks, for which the help gives no equation and no example output. (The Bruggeman and effective-medium layers, listed here until 4 October 2026, became native geometry that day: T37.)

**Physics covered.** Causal Lorentzian and Pearson line shapes.

**Procedure.** None.

**Result.** Not implemented.

**Verdict.** Not achieved: the roadmap items that the ledger makes visible.

*Where to look:* Roadmap.

### T31 · The FOCUS expert's everyday project

**Grade E · blocked ·** class I (FOCUS's numerical output) · 2026-10-04 · difes 0.16.0 · log L060

**Objective.** Opened in difes in front of its author, does one of Iñigo González de Arrieta's everyday FOCUS projects compute the same curves and refit to the same parameters?

**Reference.** An OSML export with declarations, FOCUS's computed curves and the measured data, requested in the brief.

**Data.** Not yet received.

**Physics covered.** Whatever the project holds.

**Procedure.** The one of T11 and T16: evaluate the exported definitions on FOCUS's grid, compare with FOCUS's curves, refit the data from a perturbed start.

**Result.** Pending.

**Verdict.** Not yet run: the demonstration that counts most, and the brief's first request.

*Where to look:* The brief for Iñigo (PDF).

### T32 · The variational dielectric function, RefFIT's method, on a known case

**Grade A · exact ·** class IV (physics alone) · 2026-10-04 · difes 0.17.0 · log L067

**Objective.** Does difes's implementation of Kuzmenko's variational dielectric function recover a known  $\varepsilon$  from reflectivity alone, with RefFIT's two-step recipe, and is its basis causal to the digits?

**Reference.** The construction of the RefFIT manual (§2.2.4, eqs. 2-20 to 2-22; Kuzmenko, Rev. Sci. Instrum. 76, 083108, 2005): triangular  $\varepsilon''$  on an anchor mesh with the exact Kramers–Kronig partner of each triangle; a synthetic dielectric function as the truth; the plain Kramers–Kronig phase on the same range as the conventional alternative.

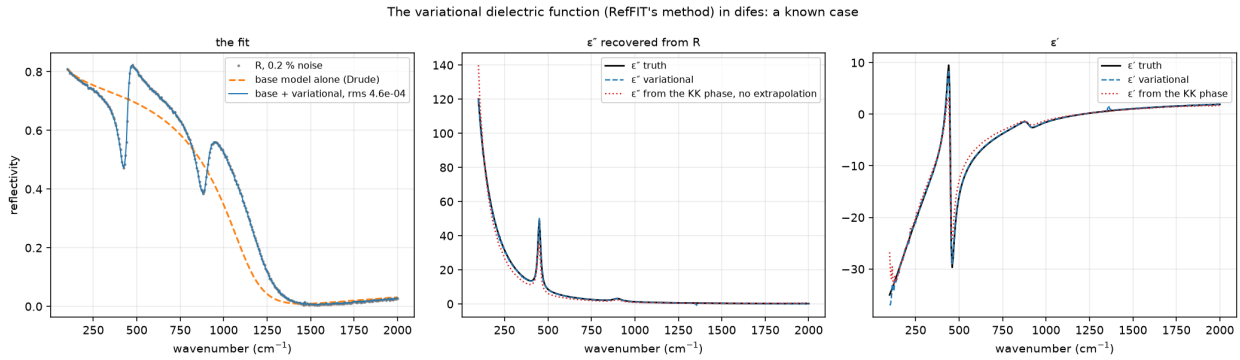
**Data.** Synthetic: the reflectivity of  $\varepsilon_\infty = 3$  plus two Lorentz oscillators (450 and 900  $\text{cm}^{-1}$ ) and a Drude term on 100–2000  $\text{cm}^{-1}$ , 381 points, with 0.2 % noise; the base model is the Drude term alone, the phonons left out on purpose.

**Physics covered.** Causality of a piecewise-linear  $\varepsilon''$  in closed form; the KK-constrained function with  $N - 2$  amplitudes and the unconstrained one with  $2N$ ; the two-step recipe (formula model frozen, variational correction added).

**Procedure.** The real part of one asymmetric triangle against the numerical Kramers–Kronig transform of its imaginary part (tests); the amplitudes fitted by the ordinary fitter with the analytic Jacobian the term provides; the recovered  $\varepsilon$  against the truth inside the range and against the plain KK phase without extrapolation; the unconstrained form fitted to  $\varepsilon'$  and  $\varepsilon''$  directly.

**Result.** The triangle’s real part equals the numerical transform to  $2 \times 10^{-3}$  of its scale on a grid to 40 000  $\text{cm}^{-1}$ ; the KK-constrained fit reaches rms  $4.6 \times 10^{-4}$  in R with 379 amplitudes and recovers  $\varepsilon''$  to 2.4 on a scale of 120 (2 %) and  $\varepsilon'$  to 1.1 inside the range, where the plain KK phase on the finite range errs by 13.9; the unconstrained form reproduces  $\varepsilon'$  and  $\varepsilon''$  to  $10^{-6}$ .

**Verdict.** Achieved against physics: the method is in difes and does what the manual says it does. The exact test, RefFIT’s own variational  $\varepsilon$  exported for the same data, waits for a RefFIT user (T34).



**Figure 23:** Left: the noisy reflectivity, the base model alone and the base plus the variational function. Middle and right:  $\varepsilon''$  and  $\varepsilon'$  recovered from R against the truth, and the plain Kramers–Kronig phase on the same finite range for comparison.

Where to look: Kramers–Kronig on a finite range (theory), the variational dielectric function; difes/src/difes/vdf.py; tests/test\_vdf.py; examples/refit/vdf\_demo.py.

## T33 · RefFIT’s model table and its line shapes

**Grade A** · **exact** · class IV (physics alone) · 2026-10-04 · difes 0.17.0 · log L067

**Objective.** Does a RefFIT model, Einf plus rows of (Wo, Wp, G) with the special codes, map onto a difes model term by term, and are RefFIT’s two extra line shapes implemented correctly?

**Reference.** The RefFIT manual’s Table 4-3 (the codes and their formulas), its Fig. 4-5 model, the Tauc–Lorentz closed form of Jellison and Modine (1996, with the erratum) checked against a numerical Kramers–Kronig transform, the Fano shape’s reduction to a Lorentzian at  $q = 0$ , and the extended-Drude outputs of §4.7.6 on a pure Drude term.

**Data.** Synthetic tables and dielectric functions; RefFIT’s manual.

**Physics covered.** Drude–Lorentz rows; signed-weight Drude and Lorentz terms; the constant term; the difference of two Lorentzians; Tauc–Lorentz; the Fano-shaped Lorentz; the factorised TO–LO formula (difes’s four-parameter mode); RefFIT’s quantities R, E1, E2, S1, S2, N1, N2, PD, LF, AC;  $1/\tau(\omega)$  and  $m^*/m$ .

**Procedure.** 16 tests: ordinary rows become Drude and Lorentz terms with the same  $\varepsilon$ ; each handled code gives the formula of the table; a missing second row and an unsupported code raise with the code named; the TO–LO rows pair into four-parameter modes and back; Fano equals Lorentz at  $q = 0$ , is causal, and its analytic derivatives match finite differences; Tauc–Lorentz’s closed-form  $\varepsilon'$  is the numerical transform of its

$\varepsilon''$ ;  $\sigma$  in practical units is  $\omega\varepsilon''/60$ ; the extended-Drude outputs of a pure Drude are  $\gamma$  and 1 at every frequency; PAR.DAT lines parse into models; the protocol files are written; the Fano term round-trips through OSML.

**Result.** All pass; the Tauc–Lorentz real part agrees with the numerical transform to  $10^{-7}$  once the gap point is treated analytically.

**Verdict.** Achieved: RefFIT’s formulas are in difes as the manual states them; what is not implemented (cuprate formulas, gapped Drude, power law, electron–boson, Tanguy) is named when met.

*Where to look:* Models of the dielectric function (theory): Fano, Tauc–Lorentz; Exchanging with RefFIT (manual); difes/src/difes/refit.py; tests/test\_refit.py.

## T34 · RefFIT’s own output: the comparison protocol, waiting for a RefFIT user

**Grade E · blocked · class I** (FOCUS’s numerical output) · 2026-10-04 · difes 0.17.0 · log L067

**Objective.** Given the same model and the same data, do RefFIT’s computed curves, its fitted parameters and its variational  $\varepsilon$  agree with difes’s?

**Reference.** RefFIT’s exported curves (ExportModelCurve), the ASC listing and the PAR.DAT line it writes on every save, produced on a RefFIT machine by the macro difes writes.

**Data.** Not yet received. The protocol (difes refit protocol) writes the dataset file, the RFS macro that sets the model row by row, exports R, E1 and E2 and saves the model; a saved RFM file is accepted instead.

**Physics covered.** Whatever the project holds; first candidates are Iñigo’s and Irati’s RefFIT projects and the manual’s own tutorial examples.

**Procedure.** Run the macro in RefFIT; read the exported curves with difes; compare with difes’s evaluation of the same table on the same grid; refit the dataset from a perturbed start and compare the parameters and their errors with the ASC listing.

**Result.** Pending.

**Verdict.** Not yet run: the class I test for RefFIT, the counterpart of what FOCUS’s exports gave. Whether SetModelParam creates table rows in a macro is also to be confirmed there.

*Where to look:* Exchanging with RefFIT (manual).

## T35 · Uncertainties following the GUM: the stated intervals cover the truth

**Grade A · exact · class IV** (physics alone) · 2026-10-04 · difes 0.18.0 · log L068

**Objective.** Does difes’s uncertainty budget, Type A plus correlated Type B components propagated by the law of propagation and by Monte Carlo, give parameter intervals that mean what they say?

**Reference.** The GUM (JCGM 100:2008) and its Supplement 1 (JCGM 101:2008); synthetic reflectivities from a known model as the truth; the fitter’s own unscaled errors as the Type A limit.

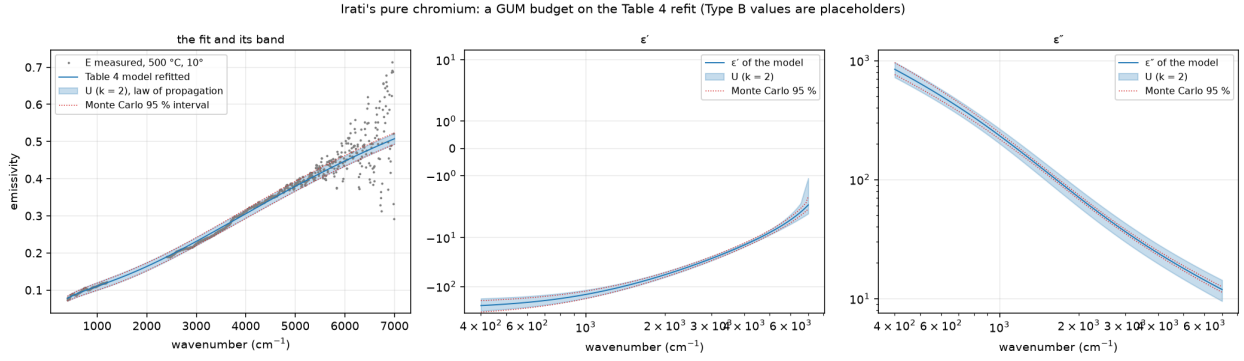
**Data.** Synthetic: the reflectivity of  $\varepsilon_\infty$  3 plus one Lorentz oscillator on 100–2000  $\text{cm}^{-1}$ , 191 points, with a 0.3 % scatter and, in the coverage test, a 1 % scale error drawn anew for each of 120 spectra; Irati’s pure chromium at 500 °C as the worked example with placeholder Type B values.

**Physics covered.** The measurement model of a least-squares fit; correlated input components (scale, offset, wavenumber shift, custom vectors); the sensitivity matrix at the optimum; Welch–Satterthwaite; derived quantities and curve bands; the Monte Carlo propagation of distributions.

**Procedure.** Seven tests: an empty budget reproduces the fitter’s unscaled Type A errors; a rectangular component gets  $u = a/\sqrt{3}$  and bad components are refused; a 1 % scale moves  $\varepsilon_\infty$  and  $\omega_p$  together, more than the scatter does, and leaves  $\omega_0$  alone; the Monte Carlo spread agrees with the law of propagation within 30 %; a derived scalar and a derived curve carry uncertainties and Monte Carlo intervals, the band widest at the peak; a poor fit raises the note and the t-factor option works; 120 synthetic fits with their  $k = 2$  intervals.

**Result.** All pass. The  $k = 2$  intervals contain the truth in at least 88 % of the trials for every parameter (the tolerance of 120 trials around 95 %). On the chromium, the budget shows the scale and offset components dominating every parameter, correlations near  $\pm 1$  between the two Drude terms, and a reduced  $\chi^2$  above 200 that the report flags: the model does not describe the data within their stated uncertainties.

**Verdict.** Achieved against physics and against the standard’s own definition of coverage. The values of the laboratory’s Type B components are still to be stated by the group; the machinery is ready for them.



**Figure 24:** Irati’s pure chromium: the refitted Table 4 model with its expanded-uncertainty band and Monte Carlo interval, and the bands on  $\varepsilon'$  and  $\varepsilon''$ , from a budget with placeholder Type B values.

*Where to look:* Uncertainties, following the GUM (theory); `difes/src/difes/uncertainty.py`; `tests/test_uncertainty.py`; `examples/uncertainty/irati_cr_budget.py`.

## T36 · The budget through the Kramers–Kronig inversion, and in a session

**Grade A** · **exact** · class IV (physics alone) · 2026-10-04 · difes 0.19.0 · log L069

**Objective.** Does the input budget propagate through a transform without a Jacobian, the finite-range Kramers–Kronig inversion, into bands on  $\varepsilon$  that behave as they should, and does the budget live in a session as a node the Bench can build?

**Reference.** The GUM Supplement 1 Monte Carlo on a known dielectric function; the session graph’s own contract (outputs depend only on the spec and its inputs).

**Data.** Synthetic: the reflectivity of a Lorentz oscillator with  $\varepsilon_\infty = 1$  on 50–5000  $\text{cm}^{-1}$  (496 points, error bars 0.002) inverted with the finite-range phase transform; a synthetic fit in a session with a 1 % scale component and 30 Monte Carlo trials.

**Physics covered.** Perturbation of a spectrum by every component (scale and offset exactly, shift through the slope, scatter from the error bars); the phase transform and  $\varepsilon$  from  $R$  and  $\theta$ ; bands as standard deviation and percentiles over the trials.

**Procedure.** Three tests: the budget serialises and deserialises with a rectangular and a digitisation component; the  $\varepsilon''$  band of the inversion grows more than twofold when the scale uncertainty goes from 0.5 % to 2 % and the 0.5 % band covers the true  $\varepsilon''$  over most of the range; in a session, the Bench’s uncertainty verb adds the node on a fit, its band spectra are bounded in  $[0, 1]$  for a reflectivity, the plot shows them, the summary lists the result, and the session round-trips through JSON with the node recomputed.

**Result.** All pass; the inversion’s coverage of the true  $\varepsilon''$  is above 60 % with the plain transform, the rest being its truncation bias, which the bands do not and should not contain.

**Verdict.** Achieved. The budget now reaches everything a session computes from a spectrum; the CFRC and the variational function go through the same function when their records are written.

*Where to look:* Uncertainties, following the GUM (theory): through an inversion; The Bench (manual): the Uncertainty card; `difes/src/difes/uncertainty.py`, `graph.py`, `bench.py`; `tests/test_uncertainty.py`.

## T37 · Effective-medium layers as native geometry: Irati’s final model from native pieces

**Grade A** · **exact** · class I (FOCUS’s numerical output) · 2026-10-04 · difes 0.20.0 · log L070

**Objective.** Built from difes’s own pieces, a metal, an anisotropic oxide as one model, a Bruggeman layer as a model with a fill fraction, and a two-layer incoherent stack, does Irati’s final Azkena model compute the reflectance FOCUS exported for it?

**Reference.** FOCUS’s exported reflectance of the definition R PhysRev doitzeko of Azkena.sml (512 points on  $0\text{--}1500\text{ cm}^{-1}$ , seven significant digits), and FOCUS’s own Eeff-BG symbol for the Bruggeman function.

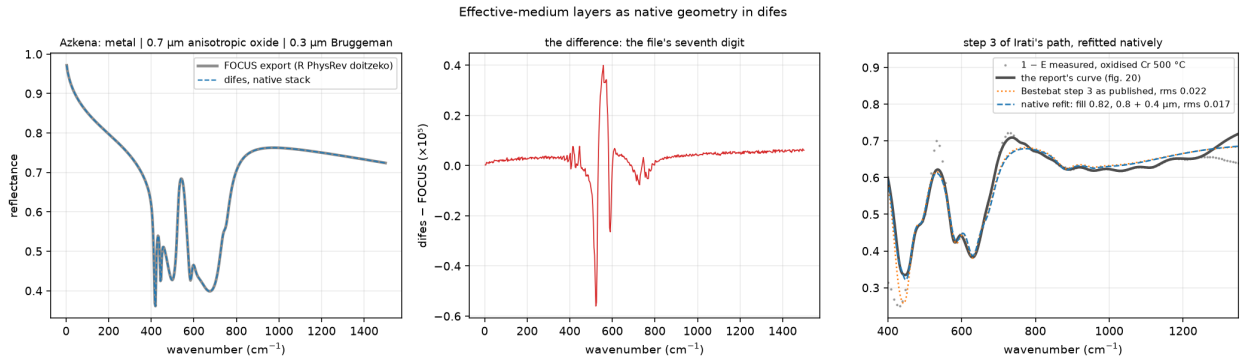
**Data.** The Azkena definitions as printed by the OSML reader: the metal ( $\varepsilon_\infty$  4.85859 and two Drude terms), the oxide as  $0.33 \times$  parallel plus  $0.66 \times$  perpendicular Lorentz sums with their own  $\varepsilon_\infty$ , a Bruggeman mixture of air and oxide at 0.5, thicknesses  $0.7\text{ }\mu\text{m}$  (oxide) and  $0.3\text{ }\mu\text{m}$  (layer); Bestebat’s step 3 ( $0.8\text{ }\mu\text{m}$  oxide under  $0.5\text{ }\mu\text{m}$  at 50 %) and the report’s drawn curve for the native refit.

**Physics covered.** Two-component Bruggeman and Maxwell–Garnett effective media; the incoherent layer stack in FOCUS’s nested form (Reflectance-Ei-E1-R12 over Reflectance-Ei-E1-E2) and in the energy-conserving form; the coherent Airy recursion; the anisotropic average written as one model with scaled strengths.

**Procedure.** Six tests: the Bruggeman limits (host at  $f = 0$ , inclusion at  $f = 1$ ), passivity and the quadratic satisfied; difes’s bruggeman against FOCUS’s Eeff-BG evaluated from the project’s own formula; a one-layer stack equal to the film in both forms and coherently; the effective-medium model’s JSON round trip and cloning; the native stack against FOCUS’s export, the same through the session’s sample node, the residual over the fill fraction with its minimum at FOCUS’s 0.5, and the fraction refitted from 0.4 on FOCUS’s curve. The example refits Bestebat’s step 3 natively against the report’s curve with the fraction free and the thicknesses scanned.

**Result.** Native stack against FOCUS’s export:  $\max |\Delta R| 5.6 \times 10^{-6}$ ,  $\text{rms } 8.6 \times 10^{-7}$ , the file’s rounding; through the session the same. The fill fraction comes back to 0.500 within  $10^{-3}$  from 0.4 (a second, shallower minimum sits near 0.22,  $\text{rms } 0.033$  against 0 at 0.5). Bestebat’s step 3 as published sits 0.022 from the report’s curve; refitted natively it reaches 0.017 with the layer at  $0.4\text{ }\mu\text{m}$  and a fraction of 0.82, a different minimum of the same construction.

**Verdict.** Achieved, exactly: what FOCUS’s formulas did inside difes for Irati’s layers is now difes’s own geometry, with the fraction as a fit parameter. A cell of the coverage matrix that read ‘not in difes yet’ this morning.



**Figure 25:** Left: FOCUS’s exported reflectance of the final Azkena model and difes’s native stack; middle: the difference times  $10^5$ ; right: Bestebat’s step 3 as published and refitted natively against the report’s curve.

*Where to look:* From  $\varepsilon$  to the measured spectrum (theory): effective-medium layers and layer stacks; Irati’s path, step by step; difes/src/difes/optics.py, models.py (EffectiveMediumModel); tests/test\_stack.py; examples/stack/irati\_native\_stack.py.

## T38 · The budget through the CFRC and the variational function

Grade A · exact · class IV (physics alone) · 2026-10-04 · difes 0.21.0 · log L071

**Objective.** Does the input budget propagate through the two corrected inversions difes offers, the CFRC and ReFIT’s variational function, and does an anchor do to the band what an anchor should?

**Reference.** The GUM Supplement 1 Monte Carlo; the definition of an anchor: a frequency where the phase is known.

**Data.** Synthetic: the reflectivity of a Lorentz oscillator with  $\varepsilon_\infty$  2.25 on 50–5000  $\text{cm}^{-1}$  (331 points), a point scatter of 0.003 as the budget; one anchor at 3000  $\text{cm}^{-1}$  with the true phase; the same spectrum in a session inverted with the Bench and the budget put on the inversion.

**Physics covered.** The finite-range phase transform; the CFRC with the two-pole phase model fitted to the anchor on every draw; the variational fit repeated on every draw; percentiles over the trials as the band.

**Procedure.** One test in four parts: the plain and the anchored inversion under the same budget, the phase band at the anchor compared; the variational function on the cropped spectrum with a sparse mesh, eight trials, bands finite and no failure; the uncertainty node placed by the Bench on an inversion node, its outputs ( $\theta$ ,  $\varepsilon'$ ,  $\varepsilon''$ ,  $n$ ,  $k$ , each with standard deviation and interval) checked and shown on the plot; the session round-tripped through JSON with the node.

**Result.** At the anchor the phase band of the corrected inversion is less than half that of the plain one; the variational bands are finite with no failed trial; the node's outputs are spectra on the data's grid with the upper band above the lower everywhere; the session reopens with the node recomputed.

**Verdict.** Achieved. Every route from a spectrum to  $\varepsilon$  that difes offers now carries the measurement's budget: the fit (T35), the plain inversion (T36), the CFRC and the variational function (T38).

*Where to look:* Uncertainties, following the GUM (theory): through an inversion; difes/src/difes/uncertainty.py (kk\_eps\_uncertainty, vdf\_uncertainty), graph.py (\_\_uncertainty\_of\_transform); tests/test\_uncertainty.py.

## T39 · The digitiser's precision measured, and made the digitisation component of the budget

**Grade A** · **exact** · class IV (physics alone) · 2026-10-04 · difes 0.22.0 · log L072

**Objective.** How precisely does the bitmap digitiser read a published figure, as a function of the figure's pixel width and of whether the curve is a line or a set of markers, so that every digitised comparison carries a measured uncertainty instead of a quoted one?

**Reference.** Known curves: a two-oscillator reflectivity drawn as a figure with matplotlib at 600, 900, 1500 and 2200 pixels of width, as lines of 1.5 and 3 pixels and as markers, read back with difes.digitize.extract\_bitmap and compared with the function that drew them.

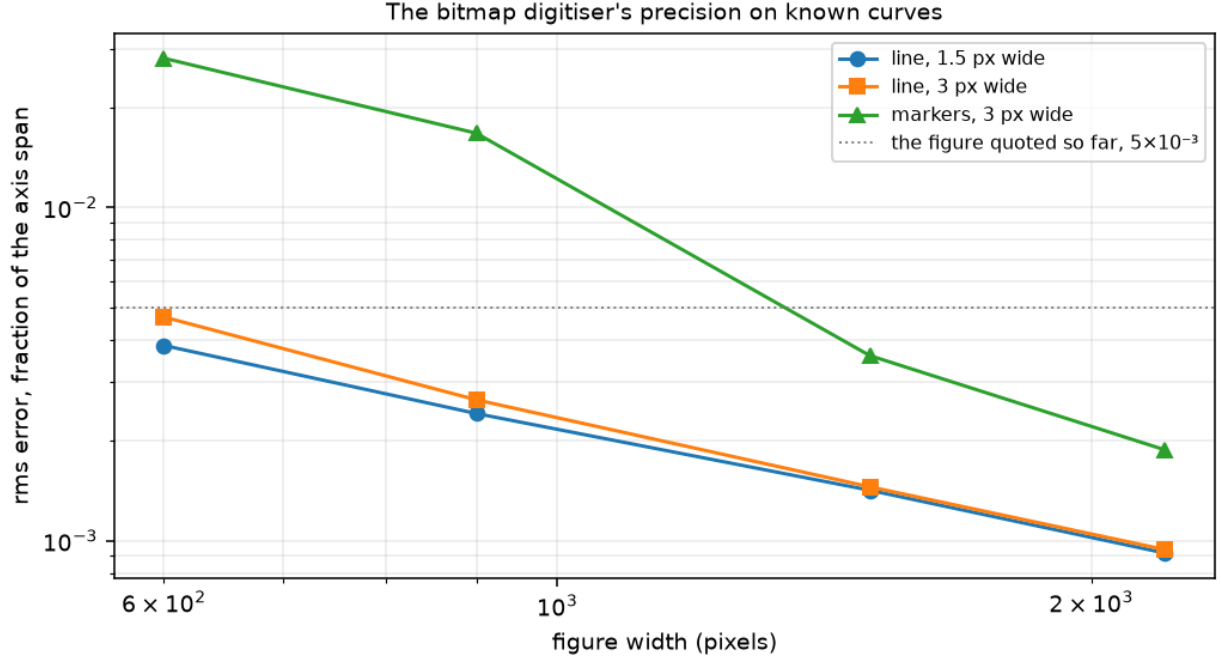
**Data.** Twelve synthetic figures; the pixel widths are those of the figures the ledger digitised (the pBN paper's Fig. 3 at 900, papers 16 and 11 at 550 and 591, the TFM's Fig. 4.11 at 2200).

**Physics covered.** None; the metrology of the method: the error in units of the axis span, its scaling with resolution, its bias, and the failure mode of markers that touch.

**Procedure.** Draw, read, interpolate the reading onto the drawing's grid, take rms, maximum and mean of the difference away from the frame; one case in the suite at 900 pixels.

**Result.** Lines: rms  $3.9 \times 10^{-3}$  of the span at 600 pixels,  $2.4 \times 10^{-3}$  at 900,  $1.4 \times 10^{-3}$  at 1500,  $0.9 \times 10^{-3}$  at 2200, nearly independent of the line width, with no bias beyond  $3 \times 10^{-4}$  and maxima of 1 to 2 % on the steep edges (the median of a column on a slope). Markers:  $2.8 \times 10^{-2}$  at 600 and  $1.7 \times 10^{-2}$  at 900 pixels, where dense markers merge and most are lost,  $3.6 \times 10^{-3}$  at 1500 and  $1.9 \times 10^{-3}$  at 2200.

**Verdict.** Achieved. The value quoted so far, about  $5 \times 10^{-3}$ , was conservative for lines and optimistic for markers at low resolution; the records T10, T21, T22 and T25 now carry the measured numbers, and Component.digitised(precision, span) puts them into a budget.



**Figure 26:** The rms error of the bitmap digitiser on known curves against the figure's width in pixels, for lines of two widths and for markers; the dotted line is the value quoted before the measurement.

Where to look: Digitising figures (manual): precision; difes/examples/digitize/precision.py; tests/test\_digitize.py.

## T40 · Iñigo's TFG (2015): RefFIT's Drude–Lorentz fits of a cBN tool, from the printed tables

**Grade B · within the data · class III** (printed tables and bitmap figures) · 2026-10-04 · difes 0.22.0 · log L073

**Objective.** Do the Drude–Lorentz parameters RefFIT fitted and the TFG printed regenerate the curves RefFIT drew, and does difes's refit of the same data land where RefFIT did? The first RefFIT-made artefact of the group's FOCUS expert, and the first record of the RefFIT branch of the ledger.

**Reference.** The TFG's Tables 2.2 (pure cBN from the literature:  $\varepsilon_\infty$  4.51, one Lorentz oscillator), 2.3 and 2.4 (the tool at 400 °C:  $\varepsilon_\infty$  11.2, Lorentz plus Drude) and 2.5 and 2.6 (1002 °C), and its Figs. 2.10 to 2.12, RefFIT's own screenshots of the fits, 376 pixels wide.

**Data.** Reflectivity  $R = 1 - E$  of the tool (black markers) and RefFIT's fitted curve (red line) read from the three screenshots with the axes calibrated on the positions of the tick labels; 240–280 data points and about 300 curve points per figure; digitisation precision at this width about  $5 \times 10^{-3}$  of the span per curve (T39), the sharp cBN line of  $7.9 \text{ cm}^{-1}$  width not resolvable at  $2.7 \text{ cm}^{-1}$  per pixel.

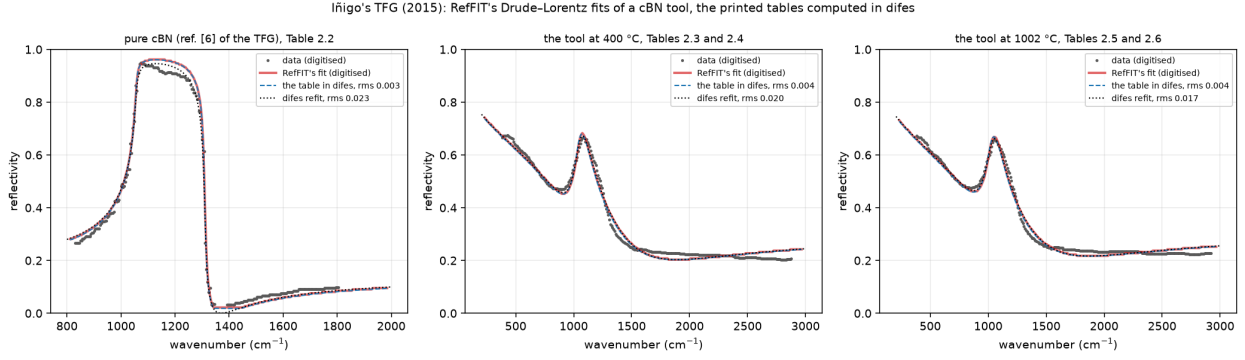
**Physics covered.**  $\varepsilon_\infty$  plus a Lorentz oscillator, plus a Drude term for the tool; bulk reflectivity at normal incidence; RefFIT's convention is difes's.

**Procedure.** Each table computed forward in difes on the curve's grid and compared with RefFIT's drawn fit and with the data; each table refitted to the digitised data with every parameter free (examples/papers/inigo\_tfg/tfg\_cbn.py).

**Result.** The tables regenerate RefFIT's curves to rms 0.0035, 0.0039 and 0.0038, the digitisation precision of the screenshots. Against the data, table, RefFIT's curve and difes's refit sit at 0.028 / 0.028 / 0.023 (pure cBN), 0.020 / 0.020 / 0.020 (400 °C) and 0.018 / 0.017 / 0.017 (1002 °C). The refits move  $\omega_0$  by 0.2 %, the plasma frequencies by 1 to 2 %, the Drude dampings by 1 to 2 %, and the Lorentz dampings by 10 % for the tool and 40 % for pure cBN, where the line is three pixels wide in the figure.

**Verdict.** Achieved within the digitisation: RefFIT's printed tables are the curves RefFIT drew, which is what the FOCUS papers of the literature campaign often failed to show, and difes refits the same data to the

same curves with the parameters within the figure’s resolution. Grade B, class III; RefFIT’s own exported curves (T34) would make it class I.



**Figure 27:** The three figures of the TFG digitised: the data, RefFIT’s fit, the printed table computed in difes and difes’s refit, for pure cBN and for the tool at 400 and 1002 °C.

*Where to look:* Exchanging with RefFIT (manual); difes/examples/papers/inigo\_tfg/tfg\_cbn.py.

## Coverage

Each capability of difes with the best grade any test reached on it. A blank is a capability with no test yet; *not in difes yet* is a roadmap item.

| capability                                                                                        | best grade          | tests                                       |
|---------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------|
| Dielectric-function profiles · Drude term                                                         | A · exact           | T01, T08, T09, T11, T27, T33, T35, T37, T40 |
| Dielectric-function profiles · Extended (double-damping) Drude                                    | C · explained       | T28                                         |
| Dielectric-function profiles · Lorentz oscillator (FOCUS phonon-3p)                               | A · exact           | T01, T08, T10, T11, T33, T35, T37, T40      |
| Dielectric-function profiles · Four-parameter factorised mode (phonon-4p)                         | A · exact           | T01, T06, T16, T23, T24, T25, T26, T28, T33 |
| Dielectric-function profiles · Anharmonic phonon, Gaussian self-energy (phonon-anharmonic)        | B · within the data | T12, T13                                    |
| Dielectric-function profiles · Anharmonic phonon with imaginary self-energy (phonon-anharmonic-i) | B · within the data | T07                                         |
| Dielectric-function profiles · Causal Gaussian band (gaussian-kkg)                                | A · exact           | T01, T12, T13, T17, T20                     |
| Dielectric-function profiles · Causal Voigt band (voigt-kkg)                                      | C · explained       | T18, T20                                    |
| Dielectric-function profiles · Voigt peak (Raman, non-causal)                                     | A · exact           | T07, T19                                    |
| Dielectric-function profiles · Cubic-spline base line                                             | A · exact           | T19                                         |
| Dielectric-function profiles · Kurosawa (LO/TO) mode                                              | B · within the data | T21, T22                                    |
| Dielectric-function profiles · Fano-shaped Lorentz oscillator (RefFIT codes –65/–66)              | A · exact           | T33                                         |
| Dielectric-function profiles · Tauc–Lorentz oscillator (RefFIT codes –55/–56)                     | A · exact           | T33                                         |

| capability                                                                                                               | best grade       | tests                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------|
| Dielectric-function profiles · lorentzian-kkg, PearsonVII, area-normalised peaks                                         | not in difes yet | T30 (blocked)                                                                                           |
| Sample optics and geometry · Bulk reflectance, normal incidence                                                          | A · exact        | T02, T08, T09, T11, T12, T13, T16, T17, T23, T24, T25, T26, T40                                         |
| Sample optics and geometry · Emissivity of an opaque sample (1 – R)                                                      | A · exact        | T02, T08, T09, T21, T27, T28                                                                            |
| Sample optics and geometry · Incoherent film on a substrate, FOCUS form (eq. 27)                                         | A · exact        | T08, T10, T11, T37                                                                                      |
| Sample optics and geometry · Incoherent film on a substrate, energy-conserving form                                      | A · exact        | T02, T08                                                                                                |
| Sample optics and geometry · Incoherent plate: R, T and E (Reflectance-E, Transmittance-E)                               | A · exact        | T02, T06, T07, T18                                                                                      |
| Sample optics and geometry · Coherent slab                                                                               | A · exact        | T02, T06                                                                                                |
| Sample optics and geometry · Fresnel coefficients at oblique incidence (FOCUS libraries)                                 | A · exact        | T06                                                                                                     |
| Sample optics and geometry · Anisotropic average $\frac{1}{3} \varepsilon_{\parallel} + \frac{2}{3} \varepsilon_{\perp}$ | A · exact        | T11, T21, T22, T37                                                                                      |
| Sample optics and geometry · Maxwell–Garnett inclusions                                                                  | A · exact        | T21, T22, T37                                                                                           |
| Sample optics and geometry · Bruggeman effective layer as a formula                                                      | A · exact        | T11, T37                                                                                                |
| Sample optics and geometry · Effective-medium layer as a native, fittable geometry; layer stacks                         | A · exact        | T37                                                                                                     |
| Fitting · Forward evaluation of a published or exported model                                                            | A · exact        | T01, T02, T04, T06, T07, T08, T11, T12, T13, T16, T17, T18, T19, T20, T22, T23, T24, T25, T32, T37, T40 |
| Fitting · Recovery of known parameters from a perturbed start                                                            | A · exact        | T04, T16, T17, T32, T35, T37                                                                            |
| Fitting · Refit of published data against the published fit                                                              | A · exact        | T09, T10, T21, T22, T24, T25, T26, T27, T28, T40                                                        |
| Fitting · Weighting by measured uncertainties or uniform                                                                 | A · exact        | T04, T09, T10, T35                                                                                      |
| Fitting · Degenerate problems recognised and stated                                                                      | A · exact        | T09, T35                                                                                                |
| Fitting · The Bench’s guided fit (strengths → positions → widths → all)                                                  | A · exact        | T05, T10                                                                                                |
| Fitting · Different minima of the same model identified                                                                  | C · explained    | T28                                                                                                     |
| Fitting · Uncertainty budget following the GUM: Type A and B, propagation, Monte Carlo, coverage                         | A · exact        | T35, T36, T38, T39                                                                                      |
| Fitting · The budget through the Kramers–Kronig inversion (Monte Carlo bands)                                            | A · exact        | T36, T38                                                                                                |
| Kramers–Kronig · KK phase of the reflectance; $\chi'$ from $\chi''$                                                      | A · exact        | T03, T32, T36                                                                                           |
| Kramers–Kronig · Multiply-subtractive KK with anchors                                                                    | A · exact        | T03                                                                                                     |
| Kramers–Kronig · Finite-range correction (CFRC)                                                                          | A · exact        | T03, T38                                                                                                |
| Kramers–Kronig · Variational dielectric function, KK-constrained and free (RefFIT’s method)                              | A · exact        | T32, T38                                                                                                |

| capability                                                                                                | best grade          | tests                                       |
|-----------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------|
| Exchange with FOCUS and RefFIT<br>· OSML model import (profiles, formulas, libraries)                     | A · exact           | T06, T07, T08, T11, T16, T17, T18, T19, T37 |
| Exchange with FOCUS and RefFIT<br>· OSML export and round trip                                            | A · exact           | T06, T33                                    |
| Exchange with FOCUS and RefFIT<br>· A whole FOCUS project as a session                                    | A · exact           | T11                                         |
| Exchange with FOCUS and RefFIT<br>· Spectra read from binary .pjt files                                   | A · exact           | T15                                         |
| Exchange with FOCUS and RefFIT<br>· RefFIT's model table (Einf, rows and codes) to and from a difes model | A · exact           | T33, T40                                    |
| Exchange with FOCUS and RefFIT<br>· RefFIT's output quantities and extended-Drude outputs                 | A · exact           | T33                                         |
| Exchange with FOCUS and RefFIT<br>· The RefFIT comparison protocol (dataset, macro, exported curves)      | no test             | T34 (blocked)                               |
| Digitising figures · Bitmap figure, colour mask and frame calibration                                     | A · exact           | T14, T21, T22, T25, T39, T40                |
| Digitising figures · Vector figure, paths read from the PDF                                               | A · exact           | T12, T13, T14, T24, T27, T28                |
| Digitising figures · Axis calibration on tick marks or tick labels                                        | B · within the data | T14, T22, T40                               |
| Software behaviour · Spectrum readers, units, conversions                                                 | A · exact           | T05, T15                                    |
| Software behaviour · The math language (parser, user functions, libraries)                                | A · exact           | T05                                         |
| Software behaviour · Session graph, views, adopt and discard                                              | A · exact           | T05, T36, T38                               |
| Software behaviour · Command line and local server                                                        | A · exact           | T05                                         |
| Software behaviour · Bench recipes and suggestions                                                        | A · exact           | T05, T36, T38                               |

## Where the evidence stands

Everything difes computes was first checked against physics, with no reference program in sight (T01 to T05). Wherever FOCUS wrote its own numbers down, difes met them to the digits of the file: Irati Tolosa's final anisotropic model, FOCUS's quartz, phosphate glass and Raman sample projects, FOCUS's own libraries and example models, four one-band exports (T06, T11, T15 to T20). Where FOCUS's result exists only as a drawn curve, difes lands on it to the width of the line, in the fluorides and sapphire of FOCUS's authors and in the FOCUS expert's own thesis (T12, T13, T27). The two places where difes and FOCUS differ are understood and quantified: FOCUS's approximation of the Voigt function, a part in a thousand of a band, and a shallow eight-parameter fit where the two programs stopped in different minima (T18, T20, T28). What remains open is between publications and their own tables, not between the programs (T22, T24 to T26); what is blocked waits for a file (T29 to T31, T34). RefFIT's variational dielectric function and its line shapes are in difes and tested against physics (T32, T33); RefFIT's own output is the next class I test. The strongest test remains T31, and it has not been run yet: one of Iñigo González de Arrieta's everyday projects, opened in difes in front of him.

*Everything in this ledger is on the project site under Tests, with the scripts, the data files and the figures.*